
Drinfeld modules

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SageMath include facilities to manipulate Drinfeld modules and their morphisms. The main entry point is the class `sage.rings.function_field.drinfeld_modules.drinfeld_module.DrinfeldModule`.

DRINFELD MODULES

1.1 Drinfeld modules

This module provides the class `sage.rings.function_field.drinfeld_module.drinfeld_module.DrinfeldModule`.

For finite Drinfeld modules and their theory of complex multiplication, see class `sage.rings.function_field.drinfeld_module.finite_drinfeld_module.DrinfeldModule`.

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- David Ayotte (2023-03): added basic j -invariants

class `sage.rings.function_field.drinfeld_modules.drinfeld_module.DrinfeldModule` (*gen, cat-e-gory*)

Bases: `Parent, UniqueRepresentation`

This class implements Drinfeld $\mathbb{F}_q[T]$ -modules.

Let $\mathbb{F}_q[T]$ be a polynomial ring with coefficients in a finite field $\mathbb{F}_q$ and let K be a field. Fix a ring morphism $\gamma : \mathbb{F}_q[T] \rightarrow K$; we say that K is an $\mathbb{F}_q[T]$ -field. Let $K\{\tau\}$ be the ring of Ore polynomials with coefficients in K , whose multiplication is given by the rule $\tau\lambda = \lambda^q\tau$ for any $\lambda \in K$.

A Drinfeld $\mathbb{F}_q[T]$ -module over the base $\mathbb{F}_q[T]$ -field K is an $\mathbb{F}_q$ -algebra morphism $\phi : \mathbb{F}_q[T] \rightarrow K\{\tau\}$ such that $\text{Im}(\phi) \not\subset K$ and ϕ agrees with γ on $\mathbb{F}_q$.

For a in $\mathbb{F}_q[T]$, $\phi(a)$ is denoted ϕ_a .

The Drinfeld $\mathbb{F}_q[T]$ -module ϕ is uniquely determined by the image ϕ_T of T ; this serves as input of the class.

Note: See also `sage.categories.drinfeld_modules`.

The *base morphism* is the morphism $\gamma : \mathbb{F}_q[T] \rightarrow K$. The monic polynomial that generates the kernel of γ is called the $\mathbb{F}_q[T]$ -characteristic, or *function-field characteristic*, of the base field. We say that $\mathbb{F}_q[T]$ is the *function ring* of ϕ ; $K\{\tau\}$ is the *Ore polynomial ring*. Further, the *generator* is ϕ_T and the *constant coefficient* is the constant coefficient of ϕ_T .

A Drinfeld module is said to be *finite* if the field K is. Despite an emphasis on this case, the base field can be any extension of $\mathbb{F}_q$:

```
sage: Fq = GF(25)
sage: A.<T> = Fq[]
sage: K.<z> = Fq.extension(6)
sage: phi = DrinfeldModule(A, [z, 4, 1])
sage: phi
Drinfeld module defined by T |--> t^2 + 4*t + z
```

```
>>> from sage.all import *
>>> Fq = GF(Integer(25))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(6), names=('z',)); (z,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [z, Integer(4), Integer(1)])
>>> phi
Drinfeld module defined by T |--> t^2 + 4*t + z
```

```
sage: Fq = GF(49)
sage: A.<T> = Fq[]
sage: K = Frac(A)
sage: psi = DrinfeldModule(A, [K(T), T+1])
sage: psi
Drinfeld module defined by T |--> (T + 1)*t + T
```

```
>>> from sage.all import *
>>> Fq = GF(Integer(49))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Frac(A)
>>> psi = DrinfeldModule(A, [K(T), T+Integer(1)])
>>> psi
Drinfeld module defined by T |--> (T + 1)*t + T
```

Note: Finite Drinfeld modules are implemented in the class `sage.rings.function_field.drinfeld_modules.finite_drinfeld_module`.

Classical references on Drinfeld modules include [Gos1998], [Rosen2002], [VS06] and [Gek1991].

Note: Drinfeld modules are defined in a larger setting, in which the polynomial ring $\mathbb{F}_q[T]$ is replaced by a more general function ring: the ring of functions in k that are regular outside ∞ , where k is a function field over $\mathbb{F}_q$ with transcendence degree 1 and ∞ is a fixed place of k . This is out of the scope of this implementation.

INPUT:

- `function_ring` – a univariate polynomial ring whose base field is a finite field
- `gen` – the generator of the Drinfeld module; as a list of coefficients or an Ore polynomial
- `name` (default: `'t'`) – the name of the Ore polynomial ring generator

Construction

A Drinfeld module object is constructed by giving the function ring and the generator:

```
sage: Fq.<z2> = GF(3^2)
sage: A.<T> = Fq[]
sage: K.<z> = Fq.extension(6)
sage: phi = DrinfeldModule(A, [z, 1, 1])
sage: phi
Drinfeld module defined by T |--> t^2 + t + z
```

```
>>> from sage.all import *
>>> Fq = GF(Integer(3)**Integer(2), names=('z2',)); (z2,) = Fq._first_ngens(1)
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(6), names=('z',)); (z,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [z, Integer(1), Integer(1)])
>>> phi
Drinfeld module defined by T |--> t^2 + t + z
```

Note: Note that the definition of the base field is implicit; it is automatically defined as the compositum of all the parents of the coefficients.

The above Drinfeld module is finite; it can also be infinite:

```
sage: L = Frac(A)
sage: psi = DrinfeldModule(A, [L(T), 1, T^3 + T + 1])
sage: psi
Drinfeld module defined by T |--> (T^3 + T + 1)*t^2 + t + T
```

```
>>> from sage.all import *
>>> L = Frac(A)
>>> psi = DrinfeldModule(A, [L(T), Integer(1), T**Integer(3) + T + Integer(1)])
>>> psi
Drinfeld module defined by T |--> (T^3 + T + 1)*t^2 + t + T
```

```
sage: phi.is_finite()
True
sage: psi.is_finite()
False
```

```
>>> from sage.all import *
>>> phi.is_finite()
True
>>> psi.is_finite()
False
```

In those examples, we used a list of coefficients ($[z, 1, 1]$) to represent the generator $\phi_T = z + t + t^2$. One can also use regular Ore polynomials:

```
sage: ore_polring = phi.ore_polring()
sage: t = ore_polring.gen()
sage: rho_T = z + t^3
sage: rho = DrinfeldModule(A, rho_T)
sage: rho
```

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```
Drinfeld module defined by  $T \mapsto t^3 + z$ 
sage: rho(T) == rho_T
True
```

```
>>> from sage.all import *
>>> ore_polring = phi.ore_polring()
>>> t = ore_polring.gen()
>>> rho_T = z + t**Integer(3)
>>> rho = DrinfeldModule(A, rho_T)
>>> rho
Drinfeld module defined by  $T \mapsto t^3 + z$ 
>>> rho(T) == rho_T
True
```

Images under the Drinfeld module are computed by calling the object:

```
sage: phi(T) # phi_T, the generator of the Drinfeld module
t^2 + t + z
sage: phi(T^3 + T + 1) # phi_(T^3 + T + 1)
t^6 + (z^11 + z^9 + 2*z^6 + 2*z^4 + 2*z + 1)*t^4
+ (2*z^11 + 2*z^10 + z^9 + z^8 + 2*z^7 + 2*z^6 + z^5 + 2*z^3)*t^3
+ (2*z^11 + z^10 + z^9 + 2*z^7 + 2*z^6 + z^5 + z^4 + 2*z^3 + 2*z + 2)*t^2
+ (2*z^11 + 2*z^8 + 2*z^6 + z^5 + z^4 + 2*z^2)*t + z^3 + z + 1
sage: phi(1) # phi_1
1
```

```
>>> from sage.all import *
>>> phi(T) # phi_T, the generator of the Drinfeld module
t^2 + t + z
>>> phi(T**Integer(3) + T + Integer(1)) # phi_(T^3 + T + 1)
t^6 + (z^11 + z^9 + 2*z^6 + 2*z^4 + 2*z + 1)*t^4
+ (2*z^11 + 2*z^10 + z^9 + z^8 + 2*z^7 + 2*z^6 + z^5 + 2*z^3)*t^3
+ (2*z^11 + z^10 + z^9 + 2*z^7 + 2*z^6 + z^5 + z^4 + 2*z^3 + 2*z + 2)*t^2
+ (2*z^11 + 2*z^8 + 2*z^6 + z^5 + z^4 + 2*z^2)*t + z^3 + z + 1
>>> phi(Integer(1)) # phi_1
1
```

The category of Drinfeld modules

Drinfeld modules have their own category (see class `sage.categories.drinfeld_modules.DrinfeldModules`):

```
sage: phi.category()
Category of Drinfeld modules over Finite Field in z of size 3^12 over its base
sage: phi.category() is psi.category()
False
sage: phi.category() is rho.category()
True
```

```
>>> from sage.all import *
>>> phi.category()
Category of Drinfeld modules over Finite Field in z of size 3^12 over its base
>>> phi.category() is psi.category()
False
```

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```
>>> phi.category() is rho.category()
True
```

One can use the category to directly create new objects:

```
sage: cat = phi.category()
sage: cat.object([z, 0, 0, 1])
Drinfeld module defined by  $T \mapsto t^3 + z$ 
```

```
>>> from sage.all import *
>>> cat = phi.category()
>>> cat.object([z, Integer(0), Integer(0), Integer(1)])
Drinfeld module defined by  $T \mapsto t^3 + z$ 
```

The base field of a Drinfeld module

The base field of the Drinfeld module is retrieved using `base()`:

```
sage: phi.base()
Finite Field in z of size  $3^{12}$  over its base
```

```
>>> from sage.all import *
>>> phi.base()
Finite Field in z of size  $3^{12}$  over its base
```

The base morphism is retrieved using `base_morphism()`:

```
sage: phi.base_morphism()
Ring morphism:
  From: Univariate Polynomial Ring in T over Finite Field in z2 of size  $3^2$ 
  To:   Finite Field in z of size  $3^{12}$  over its base
  Defn:  $T \mapsto z$ 
```

```
>>> from sage.all import *
>>> phi.base_morphism()
Ring morphism:
  From: Univariate Polynomial Ring in T over Finite Field in z2 of size  $3^2$ 
  To:   Finite Field in z of size  $3^{12}$  over its base
  Defn:  $T \mapsto z$ 
```

Note that the base field is *not* the field K . Rather, it is a ring extension (see `sage.rings.ring_extension.RingExtension`) whose underlying ring is K and whose base is the base morphism:

```
sage: phi.base() is K
False
```

```
>>> from sage.all import *
>>> phi.base() is K
False
```

Getters

One can retrieve basic properties:

```
sage: phi.base_morphism()
Ring morphism:
  From: Univariate Polynomial Ring in T over Finite Field in z2 of size 3^2
  To:   Finite Field in z of size 3^12 over its base
  Defn: T |--> z
```

```
>>> from sage.all import *
>>> phi.base_morphism()
Ring morphism:
  From: Univariate Polynomial Ring in T over Finite Field in z2 of size 3^2
  To:   Finite Field in z of size 3^12 over its base
  Defn: T |--> z
```

```
sage: phi.ore_polring() # K{t}
Ore Polynomial Ring in t over Finite Field in z of size 3^12 over its base
twisted by Frob^2
```

```
>>> from sage.all import *
>>> phi.ore_polring() # K{t}
Ore Polynomial Ring in t over Finite Field in z of size 3^12 over its base
twisted by Frob^2
```

```
sage: phi.function_ring() # Fq[T]
Univariate Polynomial Ring in T over Finite Field in z2 of size 3^2
```

```
>>> from sage.all import *
>>> phi.function_ring() # Fq[T]
Univariate Polynomial Ring in T over Finite Field in z2 of size 3^2
```

```
sage: phi.gen() # phi_T
t^2 + t + z
sage: phi.gen() == phi(T)
True
```

```
>>> from sage.all import *
>>> phi.gen() # phi_T
t^2 + t + z
>>> phi.gen() == phi(T)
True
```

```
sage: phi.constant_coefficient() # Constant coefficient of phi_T
z
```

```
>>> from sage.all import *
>>> phi.constant_coefficient() # Constant coefficient of phi_T
z
```

```
sage: phi.morphism() # The Drinfeld module as a morphism
Ring morphism:
  From: Univariate Polynomial Ring in T over Finite Field in z2 of size 3^2
```

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```

To:   Ore Polynomial Ring in t
      over Finite Field in z of size 3^12 over its base
      twisted by Frob^2
Defn: T |--> t^2 + t + z

```

```

>>> from sage.all import *
>>> phi.morphism() # The Drinfeld module as a morphism
Ring morphism:
  From: Univariate Polynomial Ring in T over Finite Field in z2 of size 3^2
  To:   Ore Polynomial Ring in t
        over Finite Field in z of size 3^12 over its base
        twisted by Frob^2
  Defn: T |--> t^2 + t + z

```

One can compute the rank and height:

```

sage: phi.rank()
2
sage: phi.height()
1

```

```

>>> from sage.all import *
>>> phi.rank()
2
>>> phi.height()
1

```

As well as the j-invariant:

```

sage: phi.j_invariant() # j-invariant
1

```

```

>>> from sage.all import *
>>> phi.j_invariant() # j-invariant
1

```

A Drinfeld $\mathbb{F}_q[T]$ -module can be seen as an Ore polynomial with positive degree and constant coefficient $\gamma(T)$, where γ is the base morphism. This analogy is the motivation for the following methods:

```

sage: phi.coefficients()
[z, 1, 1]

```

```

>>> from sage.all import *
>>> phi.coefficients()
[z, 1, 1]

```

```

sage: phi.coefficient(1)
1

```

```

>>> from sage.all import *
>>> phi.coefficient(Integer(1))
1

```

Morphisms and isogenies

A *morphism* of Drinfeld modules $\phi \rightarrow \psi$ is an Ore polynomial $f \in K\{\tau\}$ such that $f\phi_a = \psi_a f$ for every a in the function ring. In our case, this is equivalent to $f\phi_T = \psi_T f$. An *isogeny* is a nonzero morphism.

Use the `in` syntax to test if an Ore polynomial defines a morphism:

```
sage: phi(T) in Hom(phi, phi)
True
sage: t^6 in Hom(phi, phi)
True
sage: t^5 + 2*t^3 + 1 in Hom(phi, phi)
False
sage: 1 in Hom(phi, rho)
False
sage: 1 in Hom(phi, phi)
True
sage: 0 in Hom(phi, rho)
True
```

```
>>> from sage.all import *
>>> phi(T) in Hom(phi, phi)
True
>>> t**Integer(6) in Hom(phi, phi)
True
>>> t**Integer(5) + Integer(2)*t**Integer(3) + Integer(1) in Hom(phi, phi)
False
>>> Integer(1) in Hom(phi, rho)
False
>>> Integer(1) in Hom(phi, phi)
True
>>> Integer(0) in Hom(phi, rho)
True
```

To create a SageMath object representing the morphism, call the `homset` (`hom`):

```
sage: hom = Hom(phi, phi)
sage: frobenius_endomorphism = hom(t^6)
sage: identity_morphism = hom(1)
sage: zero_morphism = hom(0)
sage: frobenius_endomorphism
Endomorphism of Drinfeld module defined by T |--> t^2 + t + z
Defn: t^6
sage: identity_morphism
Identity morphism of Drinfeld module defined by T |--> t^2 + t + z
sage: zero_morphism
Endomorphism of Drinfeld module defined by T |--> t^2 + t + z
Defn: 0
```

```
>>> from sage.all import *
>>> hom = Hom(phi, phi)
>>> frobenius_endomorphism = hom(t**Integer(6))
>>> identity_morphism = hom(Integer(1))
>>> zero_morphism = hom(Integer(0))
>>> frobenius_endomorphism
Endomorphism of Drinfeld module defined by T |--> t^2 + t + z
Defn: t^6
```

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```
>>> identity_morphism
Identity morphism of Drinfeld module defined by T |--> t^2 + t + z
>>> zero_morphism
Endomorphism of Drinfeld module defined by T |--> t^2 + t + z
Defn: 0
```

The underlying Ore polynomial is retrieved with the method `ore_polynomial()`:

```
sage: frobenius_endomorphism.ore_polynomial()
t^6
sage: identity_morphism.ore_polynomial()
1
```

```
>>> from sage.all import *
>>> frobenius_endomorphism.ore_polynomial()
t^6
>>> identity_morphism.ore_polynomial()
1
```

One checks if a morphism is an isogeny, endomorphism or isomorphism:

```
sage: frobenius_endomorphism.is_isogeny()
True
sage: identity_morphism.is_isogeny()
True
sage: zero_morphism.is_isogeny()
False
sage: frobenius_endomorphism.is_isomorphism()
False
sage: identity_morphism.is_isomorphism()
True
sage: zero_morphism.is_isomorphism()
False
```

```
>>> from sage.all import *
>>> frobenius_endomorphism.is_isogeny()
True
>>> identity_morphism.is_isogeny()
True
>>> zero_morphism.is_isogeny()
False
>>> frobenius_endomorphism.is_isomorphism()
False
>>> identity_morphism.is_isomorphism()
True
>>> zero_morphism.is_isomorphism()
False
```

The Vélú formula

Let P be a nonzero Ore polynomial. We can decide if P defines an isogeny with a given domain and, if it does, find the codomain:

```
sage: P = (2*z^6 + z^3 + 2*z^2 + z + 2)*t + z^11 + 2*z^10 + 2*z^9 + 2*z^8 + z^7 +
↪ 2*z^6 + z^5 + z^3 + z^2 + z
sage: psi = phi.velu(P)
sage: psi
Drinfeld module defined by T |--> (2*z^11 + 2*z^9 + z^6 + 2*z^5 + 2*z^4 + 2*z^2 +
↪ 1)*t^2
+ (2*z^11 + 2*z^10 + 2*z^9 + z^8 + 2*z^7 + 2*z^6 + z^5 + 2*z^4 + 2*z^2 + 2*z)*t
↪ + z
sage: P in Hom(phi, psi)
True
sage: P * phi(T) == psi(T) * P
True
```

```
>>> from sage.all import *
>>> P = (Integer(2)*z**Integer(6) + z**Integer(3) + Integer(2)*z**Integer(2) + z
↪ + Integer(2))*t + z**Integer(11) + Integer(2)*z**Integer(10) +
↪ Integer(2)*z**Integer(9) + Integer(2)*z**Integer(8) + z**Integer(7) +
↪ Integer(2)*z**Integer(6) + z**Integer(5) + z**Integer(3) + z**Integer(2) + z
>>> psi = phi.velu(P)
>>> psi
Drinfeld module defined by T |--> (2*z^11 + 2*z^9 + z^6 + 2*z^5 + 2*z^4 + 2*z^2 +
↪ 1)*t^2
+ (2*z^11 + 2*z^10 + 2*z^9 + z^8 + 2*z^7 + 2*z^6 + z^5 + 2*z^4 + 2*z^2 + 2*z)*t
↪ + z
>>> P in Hom(phi, psi)
True
>>> P * phi(T) == psi(T) * P
True
```

If the input does not define an isogeny, an exception is raised:

```
sage: phi.velu(0)
Traceback (most recent call last):
...
ValueError: the input does not define an isogeny
sage: phi.velu(t)
Traceback (most recent call last):
...
ValueError: the input does not define an isogeny
```

```
>>> from sage.all import *
>>> phi.velu(Integer(0))
Traceback (most recent call last):
...
ValueError: the input does not define an isogeny
>>> phi.velu(t)
Traceback (most recent call last):
...
ValueError: the input does not define an isogeny
```

The action of a Drinfeld module

The $\mathbb{F}_q[T]$ -Drinfeld module ϕ induces a special left $\mathbb{F}_q[T]$ -module structure on any field extension L/K . Let $x \in L$ and a be in the function ring; the action is defined as $(a, x) \mapsto \phi_a(x)$. The method `action()` returns a `sage.rings.function_field.drinfeld_modules.action.Action` object representing the Drinfeld module action.

Note: In this implementation, L is K :

```
sage: action = phi.action()
sage: action
Action on Finite Field in z of size 3^12 over its base
induced by Drinfeld module defined by T |--> t^2 + t + z
```

```
>>> from sage.all import *
>>> action = phi.action()
>>> action
Action on Finite Field in z of size 3^12 over its base
induced by Drinfeld module defined by T |--> t^2 + t + z
```

The action on elements is computed by calling the action object:

```
sage: P = T + 1
sage: a = z
sage: action(P, a)
...
z^8 + 2*z^7 + 2*z^6 + 2*z^3 + z^2
sage: action(0, K.random_element())
0
sage: action(A.random_element(), 0)
0
```

```
>>> from sage.all import *
>>> P = T + Integer(1)
>>> a = z
>>> action(P, a)
...
z^9 + 2*z^8 + 2*z^7 + 2*z^6 + 2*z^3 + z^2
>>> action(Integer(0), K.random_element())
0
>>> action(A.random_element(), Integer(0))
0
```

Warning: The class `DrinfeldModuleAction` may be replaced later on. See issues #34833 and #34834.

`action()`

Return the action object (`sage.rings.function_field.drinfeld_modules.action.Action`) that represents the module action, on the base codomain, that is induced by the Drinfeld module.

OUTPUT: a Drinfeld module action object

EXAMPLES:

```

sage: Fq = GF(25)
sage: A.<T> = Fq[]
sage: K.<z12> = Fq.extension(6)
sage: p_root = 2*z12^11 + 2*z12^10 + z12^9 + 3*z12^8 + z12^7 + 2*z12^5 +
↪ 2*z12^4 + 3*z12^3 + z12^2 + 2*z12
sage: phi = DrinfeldModule(A, [p_root, z12^3, z12^5])
sage: action = phi.action()
sage: action
Action on Finite Field in z12 of size 5^12 over its base
induced by Drinfeld module defined by T |--> z12^5*t^2 + z12^3*t + 2*z12^11
+ 2*z12^10 + z12^9 + 3*z12^8 + z12^7 + 2*z12^5 + 2*z12^4 + 3*z12^3 + z12^2
↪ + 2*z12

```

```

>>> from sage.all import *
>>> Fq = GF(Integer(25))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(6), names=('z12',)); (z12,) = K._first_ngens(1)
>>> p_root = Integer(2)*z12**Integer(11) + Integer(2)*z12**Integer(10) +
↪ z12**Integer(9) + Integer(3)*z12**Integer(8) + z12**Integer(7) +
↪ Integer(2)*z12**Integer(5) + Integer(2)*z12**Integer(4) +
↪ Integer(3)*z12**Integer(3) + z12**Integer(2) + Integer(2)*z12
>>> phi = DrinfeldModule(A, [p_root, z12**Integer(3), z12**Integer(5)])
>>> action = phi.action()
>>> action
Action on Finite Field in z12 of size 5^12 over its base
induced by Drinfeld module defined by T |--> z12^5*t^2 + z12^3*t + 2*z12^11
+ 2*z12^10 + z12^9 + 3*z12^8 + z12^7 + 2*z12^5 + 2*z12^4 + 3*z12^3 + z12^2
↪ + 2*z12

```

The action on elements is computed as follows:

```

sage: P = T^2 + T + 1
sage: a = z12 + 1
sage: action(P, a)
3*z12^11 + 2*z12^10 + 3*z12^9 + 3*z12^7 + 4*z12^5 + z12^4 + z12^3 + 2*z12 + 1
sage: action(0, a)
0
sage: action(P, 0)
0

```

```

>>> from sage.all import *
>>> P = T**Integer(2) + T + Integer(1)
>>> a = z12 + Integer(1)
>>> action(P, a)
3*z12^11 + 2*z12^10 + 3*z12^9 + 3*z12^7 + 4*z12^5 + z12^4 + z12^3 + 2*z12 + 1
>>> action(Integer(0), a)
0
>>> action(P, Integer(0))
0

```

basic_j_invariant_parameters (*coeff_indices=None, nonzero=False*)

Return the list of basic j -invariant parameters.

See the method `j_invariant()` for definitions.

INPUT:

- `coeff_indices` (list or tuple, or `NoneType`; default: `None`) – indices of the Drinfeld module genera-

tor coefficients to be considered in the computation. If the parameter is `None` (default), all the coefficients are involved.

- `nonzero` (boolean, default: `False`) – if this flag is set to `True`, then only the parameters for which the corresponding basic j -invariant is nonzero are returned.

Warning: The usage of this method can be computationally expensive e.g. if the rank is greater than four, or if q is large. Setting the `nonzero` flag to `True` can speed up the computation considerably if the Drinfeld module generator possesses multiple zero coefficients.

EXAMPLES:

```
sage: A = GF(5) ['T']
sage: K.<T> = Frac(A)
sage: phi = DrinfeldModule(A, [T, 0, T+1, T^2 + 1])
sage: phi.basic_j_invariant_parameters()
[(1,), (31, 1)],
((1, 2), (1, 5, 1)),
((1, 2), (7, 4, 1)),
((1, 2), (8, 9, 2)),
((1, 2), (9, 14, 3)),
((1, 2), (10, 19, 4)),
((1, 2), (11, 24, 5)),
((1, 2), (12, 29, 6)),
((1, 2), (13, 3, 1)),
((1, 2), (15, 13, 3)),
((1, 2), (17, 23, 5)),
((1, 2), (19, 2, 1)),
((1, 2), (20, 7, 2)),
((1, 2), (22, 17, 4)),
((1, 2), (23, 22, 5)),
((1, 2), (25, 1, 1)),
((1, 2), (27, 11, 3)),
((1, 2), (29, 21, 5)),
((1, 2), (31, 31, 7)),
((2,), (31, 6))]
```

```
>>> from sage.all import *
>>> A = GF(Integer(5)) ['T']
>>> K = Frac(A, names=('T',)); (T,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [T, Integer(0), T+Integer(1), T**Integer(2) +
↳ Integer(1)])
>>> phi.basic_j_invariant_parameters()
[(1,), (31, 1)],
((1, 2), (1, 5, 1)),
((1, 2), (7, 4, 1)),
((1, 2), (8, 9, 2)),
((1, 2), (9, 14, 3)),
((1, 2), (10, 19, 4)),
((1, 2), (11, 24, 5)),
((1, 2), (12, 29, 6)),
((1, 2), (13, 3, 1)),
((1, 2), (15, 13, 3)),
((1, 2), (17, 23, 5)),
((1, 2), (19, 2, 1)),
((1, 2), (20, 7, 2)),
```

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```
((1, 2), (22, 17, 4)),
((1, 2), (23, 22, 5)),
((1, 2), (25, 1, 1)),
((1, 2), (27, 11, 3)),
((1, 2), (29, 21, 5)),
((1, 2), (31, 31, 7)),
((2,), (31, 6))]
```

Use the `nonzero=True` flag to display only the parameters whose j -invariant value is nonzero:

```
sage: phi.basic_j_invariant_parameters(nonzero=True)
[ ((2,), (31, 6)) ]
```

```
>>> from sage.all import *
>>> phi.basic_j_invariant_parameters(nonzero=True)
[ ((2,), (31, 6)) ]
```

One can specify the list of coefficients indices to be considered in the computation:

```
sage: A = GF(2)['T']
sage: K.<T> = Frac(A)
sage: phi = DrinfeldModule(A, [T, T, 1, T])
sage: phi.basic_j_invariant_parameters([1, 2])
[ ((1,), (7, 1)),
  ((1, 2), (1, 2, 1)),
  ((1, 2), (4, 1, 1)),
  ((1, 2), (5, 3, 2)),
  ((1, 2), (6, 5, 3)),
  ((1, 2), (7, 7, 4)),
  ((2,), (7, 3)) ]
```

```
>>> from sage.all import *
>>> A = GF(Integer(2))['T']
>>> K = Frac(A, names=('T',)); (T,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [T, T, Integer(1), T])
>>> phi.basic_j_invariant_parameters([Integer(1), Integer(2)])
[ ((1,), (7, 1)),
  ((1, 2), (1, 2, 1)),
  ((1, 2), (4, 1, 1)),
  ((1, 2), (5, 3, 2)),
  ((1, 2), (6, 5, 3)),
  ((1, 2), (7, 7, 4)),
  ((2,), (7, 3)) ]
```

basic_j_invariants (*nonzero=False*)

Return a dictionary whose keys are all the basic j -invariants parameters and values are the corresponding j -invariant.

See the method `j_invariant()` for definitions.

INPUT:

- `nonzero` (boolean, default: `False`) – if this flag is set to `True`, then only the parameters for which the corresponding basic j -invariant is nonzero are returned.

Warning: The usage of this method can be computationally expensive e.g. if the rank is greater than four, or if q is large. Setting the `nonzero` flag to `True` can speed up the computation considerably if the Drinfeld module generator possesses multiple zero coefficients.

EXAMPLES:

```
sage: Fq = GF(25)
sage: A.<T> = Fq[]
sage: K.<z12> = Fq.extension(6)
sage: p_root = 2*z12^11 + 2*z12^10 + z12^9 + 3*z12^8 + z12^7 + 2*z12^5 +
↳ 2*z12^4 + 3*z12^3 + z12^2 + 2*z12
sage: phi = DrinfeldModule(A, [p_root, z12^3, z12^5])
sage: phi.basic_j_invariants()
{((1,), (26, 1)): z12^10 + 4*z12^9 + 3*z12^8 + 2*z12^7 + 3*z12^6 + z12^5 +
↳ z12^3 + 4*z12^2 + z12 + 2}
```

```
>>> from sage.all import *
>>> Fq = GF(Integer(25))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(6), names=('z12',)); (z12,) = K._first_ngens(1)
>>> p_root = Integer(2)*z12**Integer(11) + Integer(2)*z12**Integer(10) +
↳ z12**Integer(9) + Integer(3)*z12**Integer(8) + z12**Integer(7) +
↳ Integer(2)*z12**Integer(5) + Integer(2)*z12**Integer(4) +
↳ Integer(3)*z12**Integer(3) + z12**Integer(2) + Integer(2)*z12
>>> phi = DrinfeldModule(A, [p_root, z12**Integer(3), z12**Integer(5)])
>>> phi.basic_j_invariants()
{((1,), (26, 1)): z12^10 + 4*z12^9 + 3*z12^8 + 2*z12^7 + 3*z12^6 + z12^5 +
↳ z12^3 + 4*z12^2 + z12 + 2}
```

```
sage: phi = DrinfeldModule(A, [p_root, 0, 1, z12])
sage: phi.basic_j_invariants(nonzero=True)
{((2,), (651, 26)): z12^11 + 3*z12^10 + 4*z12^9 + 3*z12^8 + z12^7 + 2*z12^6 +
↳ 3*z12^4 + 2*z12^3 + z12^2 + 4*z12}
```

```
>>> from sage.all import *
>>> phi = DrinfeldModule(A, [p_root, Integer(0), Integer(1), z12])
>>> phi.basic_j_invariants(nonzero=True)
{((2,), (651, 26)): z12^11 + 3*z12^10 + 4*z12^9 + 3*z12^8 + z12^7 + 2*z12^6 +
↳ 3*z12^4 + 2*z12^3 + z12^2 + 4*z12}
```

```
sage: A = GF(5)['T']
sage: K.<T> = Frac(A)
sage: phi = DrinfeldModule(A, [T, T + 2, T+1, 1])
sage: J_phi = phi.basic_j_invariants(); J_phi
{((1,), (31, 1)): T^31 + 2*T^30 + 2*T^26 + 4*T^25 + 2*T^6 + 4*T^5 + 4*T + 3,
((1, 2), (1, 5, 1)): T^6 + 2*T^5 + T + 2,
((1, 2), (7, 4, 1)): T^11 + 3*T^10 + T^9 + 4*T^8 + T^7 + 2*T^6 + 2*T^4 + 3*T^
↳ 3 + 2*T^2 + 3,
((1, 2), (8, 9, 2)): T^17 + 2*T^15 + T^14 + 4*T^13 + 4*T^11 + 4*T^10 + 3*T^9
↳ + 2*T^8 + 3*T^7 + 2*T^6 + 3*T^5 + 2*T^4 + 3*T^3 + 4*T^2 + 3*T + 1,
((1, 2), (9, 14, 3)): T^23 + 2*T^22 + 2*T^21 + T^19 + 4*T^18 + T^17 + 4*T^16
↳ + T^15 + 4*T^14 + 2*T^12 + 4*T^11 + 4*T^10 + 2*T^8 + 4*T^7 + 4*T^6 + 2*T^4
↳ + T^2 + 2*T + 2,
((1, 2), (10, 19, 4)): T^29 + 4*T^28 + T^27 + 4*T^26 + T^25 + 2*T^24 + 3*T^
↳ 23 + 2*T^22 + 3*T^21 + 2*T^20 + 4*T^19 + T^18 + 4*T^17 + T^16 + 4*T^15 + T^
```

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```

↪ 9 + 4*T^8 + T^7 + 4*T^6 + T^5 + 4*T^4 + T^3 + 4*T^2 + T + 4,
...
((2,), (31, 6)): T^31 + T^30 + T^26 + T^25 + T^6 + T^5 + T + 1}
sage: J_phi[((1, 2), (7, 4, 1))]
T^11 + 3*T^10 + T^9 + 4*T^8 + T^7 + 2*T^6 + 2*T^4 + 3*T^3 + 2*T^2 + 3

```

```

>>> from sage.all import *
>>> A = GF(Integer(5))['T']
>>> K = Frac(A, names=('T',)); (T,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [T, T + Integer(2), T+Integer(1), Integer(1)])
>>> J_phi = phi.basic_j_invariants(); J_phi
{((1,), (31, 1)): T^31 + 2*T^30 + 2*T^26 + 4*T^25 + 2*T^6 + 4*T^5 + 4*T + 3,
 ((1, 2), (1, 5, 1)): T^6 + 2*T^5 + T + 2,
 ((1, 2), (7, 4, 1)): T^11 + 3*T^10 + T^9 + 4*T^8 + T^7 + 2*T^6 + 2*T^4 + 3*T^
↪ 3 + 2*T^2 + 3,
 ((1, 2), (8, 9, 2)): T^17 + 2*T^15 + T^14 + 4*T^13 + 4*T^11 + 4*T^10 + 3*T^9
↪ + 2*T^8 + 3*T^7 + 2*T^6 + 3*T^5 + 2*T^4 + 3*T^3 + 4*T^2 + 3*T + 1,
 ((1, 2), (9, 14, 3)): T^23 + 2*T^22 + 2*T^21 + T^19 + 4*T^18 + T^17 + 4*T^16
↪ + T^15 + 4*T^14 + 2*T^12 + 4*T^11 + 4*T^10 + 2*T^8 + 4*T^7 + 4*T^6 + 2*T^4
↪ + T^2 + 2*T + 2,
 ((1, 2), (10, 19, 4)): T^29 + 4*T^28 + T^27 + 4*T^26 + T^25 + 2*T^24 + 3*T^
↪ 23 + 2*T^22 + 3*T^21 + 2*T^20 + 4*T^19 + T^18 + 4*T^17 + T^16 + 4*T^15 + T^
↪ 9 + 4*T^8 + T^7 + 4*T^6 + T^5 + 4*T^4 + T^3 + 4*T^2 + T + 4,
...
((2,), (31, 6)): T^31 + T^30 + T^26 + T^25 + T^6 + T^5 + T + 1}
>>> J_phi[(Integer(1), Integer(2)), (Integer(7), Integer(4), Integer(1))]
T^11 + 3*T^10 + T^9 + 4*T^8 + T^7 + 2*T^6 + 2*T^4 + 3*T^3 + 2*T^2 + 3

```

coefficient (n)

Return the n -th coefficient of the generator.

INPUT:

- n – a nonnegative integer

OUTPUT: an element in the base codomain

EXAMPLES:

```

sage: Fq = GF(25)
sage: A.<T> = Fq[]
sage: K.<z12> = Fq.extension(6)
sage: p_root = 2*z12^11 + 2*z12^10 + z12^9 + 3*z12^8 + z12^7 + 2*z12^5 +
↪ 2*z12^4 + 3*z12^3 + z12^2 + 2*z12
sage: phi = DrinfeldModule(A, [p_root, z12^3, z12^5])
sage: phi.coefficient(0)
2*z12^11 + 2*z12^10 + z12^9 + 3*z12^8 + z12^7 + 2*z12^5
+ 2*z12^4 + 3*z12^3 + z12^2 + 2*z12
sage: phi.coefficient(0) == p_root
True
sage: phi.coefficient(1)
z12^3
sage: phi.coefficient(2)
z12^5
sage: phi.coefficient(5)
Traceback (most recent call last):
...
ValueError: input must be >= 0 and <= rank

```

```

>>> from sage.all import *
>>> Fq = GF(Integer(25))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(6), names=('z12',)); (z12,) = K._first_ngens(1)
>>> p_root = Integer(2)*z12**Integer(11) + Integer(2)*z12**Integer(10) +
↳ z12**Integer(9) + Integer(3)*z12**Integer(8) + z12**Integer(7) +
↳ Integer(2)*z12**Integer(5) + Integer(2)*z12**Integer(4) +
↳ Integer(3)*z12**Integer(3) + z12**Integer(2) + Integer(2)*z12
>>> phi = DrinfeldModule(A, [p_root, z12**Integer(3), z12**Integer(5)])
>>> phi.coefficient(Integer(0))
2*z12^11 + 2*z12^10 + z12^9 + 3*z12^8 + z12^7 + 2*z12^5
+ 2*z12^4 + 3*z12^3 + z12^2 + 2*z12
>>> phi.coefficient(Integer(0)) == p_root
True
>>> phi.coefficient(Integer(1))
z12^3
>>> phi.coefficient(Integer(2))
z12^5
>>> phi.coefficient(Integer(5))
Traceback (most recent call last):
...
ValueError: input must be >= 0 and <= rank

```

coefficients (*sparse=True*)

Return the coefficients of the generator, as a list.

If the flag `sparse` is `True` (default), only return the nonzero coefficients; otherwise, return all of them.

INPUT:

- `sparse` – a boolean

EXAMPLES:

```

sage: Fq = GF(25)
sage: A.<T> = Fq[]
sage: K.<z12> = Fq.extension(6)
sage: p_root = 2*z12^11 + 2*z12^10 + z12^9 + 3*z12^8 + z12^7 + 2*z12^5 +
↳ 2*z12^4 + 3*z12^3 + z12^2 + 2*z12
sage: phi = DrinfeldModule(A, [p_root, z12^3, z12^5])
sage: phi.coefficients()
[2*z12^11 + 2*z12^10 + z12^9 + 3*z12^8 + z12^7
 + 2*z12^5 + 2*z12^4 + 3*z12^3 + z12^2 + 2*z12,
 z12^3,
 z12^5]

```

```

>>> from sage.all import *
>>> Fq = GF(Integer(25))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(6), names=('z12',)); (z12,) = K._first_ngens(1)
>>> p_root = Integer(2)*z12**Integer(11) + Integer(2)*z12**Integer(10) +
↳ z12**Integer(9) + Integer(3)*z12**Integer(8) + z12**Integer(7) +
↳ Integer(2)*z12**Integer(5) + Integer(2)*z12**Integer(4) +
↳ Integer(3)*z12**Integer(3) + z12**Integer(2) + Integer(2)*z12
>>> phi = DrinfeldModule(A, [p_root, z12**Integer(3), z12**Integer(5)])
>>> phi.coefficients()
[2*z12^11 + 2*z12^10 + z12^9 + 3*z12^8 + z12^7
 + 2*z12^5 + 2*z12^4 + 3*z12^3 + z12^2 + 2*z12,

```

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```
z12^3,
z12^5]
```

Careful, the method only returns the nonzero coefficients, unless otherwise specified:

```
sage: rho = DrinfeldModule(A, [p_root, 0, 0, 0, 1])
sage: rho.coefficients()
[2*z12^11 + 2*z12^10 + z12^9 + 3*z12^8 + z12^7
 + 2*z12^5 + 2*z12^4 + 3*z12^3 + z12^2 + 2*z12,
 1]
sage: rho.coefficients(sparse=False)
[2*z12^11 + 2*z12^10 + z12^9 + 3*z12^8 + z12^7
 + 2*z12^5 + 2*z12^4 + 3*z12^3 + z12^2 + 2*z12,
 0,
 0,
 0,
 1]
```

```
>>> from sage.all import *
>>> rho = DrinfeldModule(A, [p_root, Integer(0), Integer(0), Integer(0),
↳Integer(1)])
>>> rho.coefficients()
[2*z12^11 + 2*z12^10 + z12^9 + 3*z12^8 + z12^7
 + 2*z12^5 + 2*z12^4 + 3*z12^3 + z12^2 + 2*z12,
 1]
>>> rho.coefficients(sparse=False)
[2*z12^11 + 2*z12^10 + z12^9 + 3*z12^8 + z12^7
 + 2*z12^5 + 2*z12^4 + 3*z12^3 + z12^2 + 2*z12,
 0,
 0,
 0,
 1]
```

gen()

Return the generator of the Drinfeld module.

EXAMPLES:

```
sage: Fq = GF(25)
sage: A.<T> = Fq[]
sage: K.<z12> = Fq.extension(6)
sage: p_root = 2*z12^11 + 2*z12^10 + z12^9 + 3*z12^8 + z12^7 + 2*z12^5 +
↳2*z12^4 + 3*z12^3 + z12^2 + 2*z12
sage: phi = DrinfeldModule(A, [p_root, z12^3, z12^5])
sage: phi.gen() == phi(T)
True
```

```
>>> from sage.all import *
>>> Fq = GF(Integer(25))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(6), names=('z12',)); (z12,) = K._first_ngens(1)
>>> p_root = Integer(2)*z12**Integer(11) + Integer(2)*z12**Integer(10) +
↳z12**Integer(9) + Integer(3)*z12**Integer(8) + z12**Integer(7) +
↳Integer(2)*z12**Integer(5) + Integer(2)*z12**Integer(4) +
↳Integer(3)*z12**Integer(3) + z12**Integer(2) + Integer(2)*z12
>>> phi = DrinfeldModule(A, [p_root, z12**Integer(3), z12**Integer(5)])
```

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```
>>> phi.gen() == phi(T)
True
```

height()

Return the height of the Drinfeld module if the function field characteristic is a prime ideal; raise `ValueError` otherwise.

The height of a Drinfeld module is defined when the function field characteristic is a prime ideal. In our case, this ideal is even generated by a monic polynomial p in the function field. Write $\phi_p = a_s \tau^s + \dots + \tau^{r \cdot \deg(p)}$. The height of the Drinfeld module is the well-defined positive integer $h = \frac{s}{\deg(p)}$.

Note: See [Gos1998], Definition 4.5.8 for the general definition.

EXAMPLES:

```
sage: Fq = GF(25)
sage: A.<T> = Fq[]
sage: K.<z12> = Fq.extension(6)
sage: p_root = 2*z12^11 + 2*z12^10 + z12^9 + 3*z12^8 + z12^7 + 2*z12^5 +
↳ 2*z12^4 + 3*z12^3 + z12^2 + 2*z12
sage: phi = DrinfeldModule(A, [p_root, z12^3, z12^5])
sage: phi.height() == 1
True
sage: phi.is_ordinary()
True
```

```
>>> from sage.all import *
>>> Fq = GF(Integer(25))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(6), names=('z12',)); (z12,) = K._first_ngens(1)
>>> p_root = Integer(2)*z12**Integer(11) + Integer(2)*z12**Integer(10) +
↳ z12**Integer(9) + Integer(3)*z12**Integer(8) + z12**Integer(7) +
↳ Integer(2)*z12**Integer(5) + Integer(2)*z12**Integer(4) +
↳ Integer(3)*z12**Integer(3) + z12**Integer(2) + Integer(2)*z12
>>> phi = DrinfeldModule(A, [p_root, z12**Integer(3), z12**Integer(5)])
>>> phi.height() == Integer(1)
True
>>> phi.is_ordinary()
True
```

```
sage: Fq = GF(343)
sage: A.<T> = Fq[]
sage: K.<z6> = Fq.extension(2)
sage: phi = DrinfeldModule(A, [1, 0, z6])
sage: phi.height()
2
sage: phi.is_supersingular()
True
```

```
>>> from sage.all import *
>>> Fq = GF(Integer(343))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(2), names=('z6',)); (z6,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [Integer(1), Integer(0), z6])
```

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```
>>> phi.height()
2
>>> phi.is_supersingular()
True
```

In characteristic zero, height is not defined:

```
sage: L = A.fraction_field()
sage: phi = DrinfeldModule(A, [L(T), L(1)])
sage: phi.height()
Traceback (most recent call last):
...
ValueError: height is only defined for prime function field characteristic
```

```
>>> from sage.all import *
>>> L = A.fraction_field()
>>> phi = DrinfeldModule(A, [L(T), L(Integer(1))])
>>> phi.height()
Traceback (most recent call last):
...
ValueError: height is only defined for prime function field characteristic
```

hom (*x*, *codomain=None*)

Return the homomorphism defined by *x* having this Drinfeld module as domain.

We recall that a homomorphism $f : \phi \rightarrow \psi$ between two Drinfeld modules is defined by an Ore polynomial u , which is subject to the relation $\phi_T u = u \psi_T$.

INPUT:

- *x* – an element of the ring of functions, or an Ore polynomial
- *codomain* – a Drinfeld module or None (default: None)

EXAMPLES:

```
sage: Fq = GF(5)
sage: A.<T> = Fq[]
sage: K.<z> = Fq.extension(3)
sage: phi = DrinfeldModule(A, [z, 0, 1, z])
sage: phi
Drinfeld module defined by T |--> z*t^3 + t^2 + z
```

```
>>> from sage.all import *
>>> Fq = GF(Integer(5))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(3), names=('z',)); (z,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [z, Integer(0), Integer(1), z])
>>> phi
Drinfeld module defined by T |--> z*t^3 + t^2 + z
```

An important class of endomorphisms of a Drinfeld module ϕ is given by scalar multiplications, that are endomorphisms corresponding to the Ore polynomials ϕ_a with a in the function ring A . We construct them as follows:

```
sage: phi.hom(T)
Endomorphism of Drinfeld module defined by T |--> z*t^3 + t^2 + z
Defn: z*t^3 + t^2 + z
```

```
>>> from sage.all import *
>>> phi.hom(T)
Endomorphism of Drinfeld module defined by T |--> z*t^3 + t^2 + z
Defn: z*t^3 + t^2 + z
```

```
sage: phi.hom(T^2 + 1)
Endomorphism of Drinfeld module defined by T |--> z*t^3 + t^2 + z
Defn: z^2*t^6 + (3*z^2 + z + 1)*t^5 + t^4 + 2*z^2*t^3 + (3*z^2 + z + 1)*t^2 +
↪ z^2 + 1
```

```
>>> from sage.all import *
>>> phi.hom(T**Integer(2) + Integer(1))
Endomorphism of Drinfeld module defined by T |--> z*t^3 + t^2 + z
Defn: z^2*t^6 + (3*z^2 + z + 1)*t^5 + t^4 + 2*z^2*t^3 + (3*z^2 + z + 1)*t^2 +
↪ z^2 + 1
```

We can also define a morphism by passing in the Ore polynomial defining it. For example, below, we construct the Frobenius endomorphism of ϕ :

```
sage: t = phi.ore_variable()
sage: phi.hom(t^3)
Endomorphism of Drinfeld module defined by T |--> z*t^3 + t^2 + z
Defn: t^3
```

```
>>> from sage.all import *
>>> t = phi.ore_variable()
>>> phi.hom(t**Integer(3))
Endomorphism of Drinfeld module defined by T |--> z*t^3 + t^2 + z
Defn: t^3
```

If the input Ore polynomial defines a morphism to another Drinfeld module, the latter is determined automatically:

```
sage: phi.hom(t + 1)
Drinfeld Module morphism:
  From: Drinfeld module defined by T |--> z*t^3 + t^2 + z
  To:   Drinfeld module defined by T |--> (2*z^2 + 4*z + 4)*t^3 + (3*z^2 +
↪ 2*z + 2)*t^2 + (2*z^2 + 3*z + 4)*t + z
Defn: t + 1
```

```
>>> from sage.all import *
>>> phi.hom(t + Integer(1))
Drinfeld Module morphism:
  From: Drinfeld module defined by T |--> z*t^3 + t^2 + z
  To:   Drinfeld module defined by T |--> (2*z^2 + 4*z + 4)*t^3 + (3*z^2 +
↪ 2*z + 2)*t^2 + (2*z^2 + 3*z + 4)*t + z
Defn: t + 1
```

is_finite()

Return True if this Drinfeld module is finite, False otherwise.

EXAMPLES:

```

sage: Fq = GF(25)
sage: A.<T> = Fq[]
sage: K.<z12> = Fq.extension(6)
sage: p_root = 2*z12^11 + 2*z12^10 + z12^9 + 3*z12^8 + z12^7 + 2*z12^5 +
↳ 2*z12^4 + 3*z12^3 + z12^2 + 2*z12
sage: phi = DrinfeldModule(A, [p_root, z12^3, z12^5])
sage: phi.is_finite()
True
sage: B.<Y> = Fq[]
sage: L = Frac(B)
sage: psi = DrinfeldModule(A, [L(2), L(1)])
sage: psi.is_finite()
False

```

```

>>> from sage.all import *
>>> Fq = GF(Integer(25))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(6), names=('z12',)); (z12,) = K._first_ngens(1)
>>> p_root = Integer(2)*z12**Integer(11) + Integer(2)*z12**Integer(10) +
↳ z12**Integer(9) + Integer(3)*z12**Integer(8) + z12**Integer(7) +
↳ Integer(2)*z12**Integer(5) + Integer(2)*z12**Integer(4) +
↳ Integer(3)*z12**Integer(3) + z12**Integer(2) + Integer(2)*z12
>>> phi = DrinfeldModule(A, [p_root, z12**Integer(3), z12**Integer(5)])
>>> phi.is_finite()
True
>>> B = Fq['Y']; (Y,) = B._first_ngens(1)
>>> L = Frac(B)
>>> psi = DrinfeldModule(A, [L(Integer(2)), L(Integer(1))])
>>> psi.is_finite()
False

```

is_isomorphic (*other*, *absolutely=False*)

Return True if this Drinfeld module is isomorphic to *other*; return False otherwise.

INPUT:

- *absolutely* – a boolean (default: False); if True, check the existence of an isomorphism defined on the base field; if False, check over an algebraic closure.

EXAMPLES:

```

sage: Fq = GF(5)
sage: A.<T> = Fq[]
sage: K.<z> = Fq.extension(3)
sage: phi = DrinfeldModule(A, [z, 0, 1, z])
sage: t = phi.ore_variable()

```

```

>>> from sage.all import *
>>> Fq = GF(Integer(5))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(3), names=('z',)); (z,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [z, Integer(0), Integer(1), z])
>>> t = phi.ore_variable()

```

We create a second Drinfeld module, which is isomorphic to ϕ and then check that they are indeed isomorphic:

```
sage: psi = phi.velu(z)
sage: phi.is_isomorphic(psi)
True
```

```
>>> from sage.all import *
>>> psi = phi.velu(z)
>>> phi.is_isomorphic(psi)
True
```

In the example below, ϕ and ψ are isogenous but not isomorphic:

```
sage: psi = phi.velu(t + 1)
sage: phi.is_isomorphic(psi)
False
```

```
>>> from sage.all import *
>>> psi = phi.velu(t + Integer(1))
>>> phi.is_isomorphic(psi)
False
```

Here is an example of two Drinfeld modules which are isomorphic on an algebraic closure but not on the base field:

```
sage: phi = DrinfeldModule(A, [z, 1])
sage: psi = DrinfeldModule(A, [z, z])
sage: phi.is_isomorphic(psi)
False
sage: phi.is_isomorphic(psi, absolutely=True)
True
```

```
>>> from sage.all import *
>>> phi = DrinfeldModule(A, [z, Integer(1)])
>>> psi = DrinfeldModule(A, [z, z])
>>> phi.is_isomorphic(psi)
False
>>> phi.is_isomorphic(psi, absolutely=True)
True
```

On certain fields, testing isomorphisms over the base field may fail:

```
sage: L = A.fraction_field()
sage: T = L.gen()
sage: phi = DrinfeldModule(A, [T, 0, 1])
sage: psi = DrinfeldModule(A, [T, 0, T])
sage: psi.is_isomorphic(phi)
Traceback (most recent call last):
...
NotImplementedError: cannot solve the equation u^24 == T
```

```
>>> from sage.all import *
>>> L = A.fraction_field()
>>> T = L.gen()
>>> phi = DrinfeldModule(A, [T, Integer(0), Integer(1)])
>>> psi = DrinfeldModule(A, [T, Integer(0), T])
>>> psi.is_isomorphic(phi)
Traceback (most recent call last):
```

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```
...
NotImplementedError: cannot solve the equation u^24 == T
```

However, it never fails over the algebraic closure:

```
sage: psi.is_isomorphic(phi, absolutely=True)
True
```

```
>>> from sage.all import *
>>> psi.is_isomorphic(phi, absolutely=True)
True
```

Note finally that when the constant coefficients of ϕ_T and ψ_T differ, ϕ and ψ do not belong to the same category and checking whether they are isomorphic does not make sense; in this case, an error is raised:

```
sage: phi = DrinfeldModule(A, [z, 0, 1])
sage: psi = DrinfeldModule(A, [z^2, 0, 1])
sage: phi.is_isomorphic(psi)
Traceback (most recent call last):
...
ValueError: Drinfeld modules are not in the same category
```

```
>>> from sage.all import *
>>> phi = DrinfeldModule(A, [z, Integer(0), Integer(1)])
>>> psi = DrinfeldModule(A, [z**Integer(2), Integer(0), Integer(1)])
>>> phi.is_isomorphic(psi)
Traceback (most recent call last):
...
ValueError: Drinfeld modules are not in the same category
```

`j_invariant` (parameter=None, check=True)

Return the j -invariant of the Drinfeld $\mathbb{F}_q[T]$ -module for the given parameter.

Suppose that $\phi_T = g_0 + g_1\tau + \dots + g_r\tau^r$ with $g_r \neq 0$. Then the $((k_1, \dots, k_n), (d_1, \dots, d_n, d_r))$ - j -invariant of ϕ is defined by

$$j_{k_1, \dots, k_n}^{d_1, \dots, d_n, d_r}(\phi) := \frac{1}{g_r^{d_r}} \prod_{i=1}^n g_{k_i}^{d_i}$$

where $1 \leq k_1 < k_2 < \dots < k_n \leq r-1$ and the integers d_i satisfy the *weight-0 condition*:

$$d_1(q^{k_1} - 1) + d_2(q^{k_2} - 1) + \dots + d_n(q^{k_n} - 1) = d_r(q^r - 1).$$

Furthermore, if $\gcd(d_1, \dots, d_n, d_r) = 1$ and

$$0 \leq d_i \leq (q^r - 1)/(q^{\gcd(i, r)} - 1), \quad 1 \leq i \leq n,$$

then the j -invariant is called *basic*. See the method `basic_j_invariant_parameters()` for computing the list of all basic j -invariant parameters.

Note: In [Pap2023], Papikian follows a slightly different convention:

- His j -invariants (see Definition 3.8.7) correspond to our basic j -invariants, as defined above.
- His *basic* j -invariant (see Example 3.8.10) correspond to our j_k -invariants, as implemented in `jk_invariants()`.

We chose to follow Potemine's convention, as he introduced those objects in [Pot1998]. Theorem 2.2 of [Pot1998] or Theorem 3.8.11 of [Pap2023] assert that two Drinfeld $\mathbb{F}_q[T]$ -modules over K are isomorphic over the separable closure of K if and only if their basic j -invariants (as implemented here) coincide for any well-defined couple of tuples $((k_1, k_2, \dots, k_n), (d_1, d_2, \dots, d_n, d_r)), .$

INPUT:

- `parameter` (tuple or list, integer or `NoneType`; default: `None`) – the j -invariant parameter:
 - If `parameter` is a list or a tuple, then it must be of the form: $((k_1, k_2, \dots, k_n), (d_1, d_2, \dots, d_n, d_r))$, where the k_i and d_i are integers satisfying the weight-0 condition described above.
 - If `parameter` is an integer k then the method returns the j -invariant associated to the parameter $((k,), (d_k, d_r))$;
 - If `parameter` is `None` and the rank of the Drinfeld module is 2, then the method returns its usual j -invariant, that is the j -invariant for the parameter $((1,), (q + 1, 1))$.
- `check` (bool, default: `True`) – if this flag is set to `False` then the code will not check if the given parameter is valid and satisfy the weight-0 condition.

OUTPUT: the j -invariant of `self` for the given parameter.

EXAMPLES:

```
sage: Fq = GF(25)
sage: A.<T> = Fq[]
sage: K.<z12> = Fq.extension(6)
sage: p_root = 2*z12^11 + 2*z12^10 + z12^9 + 3*z12^8 + z12^7 + 2*z12^5 +
↪ 2*z12^4 + 3*z12^3 + z12^2 + 2*z12
sage: phi = DrinfeldModule(A, [p_root, z12^3, z12^5])
sage: phi.j_invariant()
z12^10 + 4*z12^9 + 3*z12^8 + 2*z12^7 + 3*z12^6 + z12^5 + z12^3 + 4*z12^2 +
↪ z12 + 2
sage: psi = DrinfeldModule(A, [p_root, 1, 1])
sage: psi.j_invariant()
1
sage: rho = DrinfeldModule(A, [p_root, 0, 1])
sage: rho.j_invariant()
0
```

```
>>> from sage.all import *
>>> Fq = GF(Integer(25))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(6), names=('z12',)); (z12,) = K._first_ngens(1)
>>> p_root = Integer(2)*z12**Integer(11) + Integer(2)*z12**Integer(10) +
↪ z12**Integer(9) + Integer(3)*z12**Integer(8) + z12**Integer(7) +
↪ Integer(2)*z12**Integer(5) + Integer(2)*z12**Integer(4) +
↪ Integer(3)*z12**Integer(3) + z12**Integer(2) + Integer(2)*z12
>>> phi = DrinfeldModule(A, [p_root, z12**Integer(3), z12**Integer(5)])
>>> phi.j_invariant()
z12^10 + 4*z12^9 + 3*z12^8 + 2*z12^7 + 3*z12^6 + z12^5 + z12^3 + 4*z12^2 +
↪ z12 + 2
>>> psi = DrinfeldModule(A, [p_root, Integer(1), Integer(1)])
>>> psi.j_invariant()
1
>>> rho = DrinfeldModule(A, [p_root, Integer(0), Integer(1)])
```

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```
>>> rho.j_invariant()
0
```

```
sage: A = GF(5)['T']
sage: K.<T> = Frac(A)
sage: phi = DrinfeldModule(A, [T, T^2, 1, T + 1, T^3])
sage: phi.j_invariant(1)
T^309
sage: phi.j_invariant(2)
1/T^3
sage: phi.j_invariant(3)
(T^156 + T^155 + T^151 + T^150 + T^131 + T^130 + T^126 + T^125 + T^31 + T^30
↪ + T^26 + T^25 + T^6 + T^5 + T + 1)/T^93
```

```
>>> from sage.all import *
>>> A = GF(Integer(5))['T']
>>> K = Frac(A, names=('T',)); (T,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [T, T**Integer(2), Integer(1), T + Integer(1),
↪ T**Integer(3)])
>>> phi.j_invariant(Integer(1))
T^309
>>> phi.j_invariant(Integer(2))
1/T^3
>>> phi.j_invariant(Integer(3))
(T^156 + T^155 + T^151 + T^150 + T^131 + T^130 + T^126 + T^125 + T^31 + T^30
↪ + T^26 + T^25 + T^6 + T^5 + T + 1)/T^93
```

The parameter can either be a tuple or a list:

```
sage: Fq.<a> = GF(7)
sage: A.<T> = Fq[]
sage: phi = DrinfeldModule(A, [a, a^2 + a, 0, 3*a, a^2+1])
sage: J = phi.j_invariant((1, 3), (267, 269, 39)); J
5
sage: J == (phi.coefficient(1)**267)*(phi.coefficient(3)**269)/(phi.
↪ coefficient(4)**39)
True
sage: phi.j_invariant([[3], [400, 57]])
4
sage: phi.j_invariant([[3], [400, 57]]) == phi.j_invariant(3)
True
```

```
>>> from sage.all import *
>>> Fq = GF(Integer(7), names=('a',)); (a,) = Fq._first_ngens(1)
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> phi = DrinfeldModule(A, [a, a**Integer(2) + a, Integer(0), Integer(3)*a,
↪ a**Integer(2)+Integer(1)])
>>> J = phi.j_invariant((Integer(1), Integer(3)), (Integer(267),
↪ Integer(269), Integer(39))); J
5
>>> J == (phi.coefficient(Integer(1))**Integer(267))*(phi.
↪ coefficient(Integer(3))**Integer(269))/(phi.
↪ coefficient(Integer(4))**Integer(39))
True
>>> phi.j_invariant([Integer(3)], [Integer(400), Integer(57)])
4
```

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```
>>> phi.j_invariant([[Integer(3)], [Integer(400), Integer(57)]]) == phi.j_
↪invariant(Integer(3))
True
```

The list of all basic j -invariant parameters can be retrieved using the method `basic_j_invariant_parameters()`:

```
sage: A = GF(3)['T']
sage: K.<T> = Frac(A)
sage: phi = DrinfeldModule(A, [T, T^2 + T + 1, 0, T^4 + 1, T - 1])
sage: param = phi.basic_j_invariant_parameters(nonzero=True)
sage: phi.j_invariant(param[1])
T^13 + 2*T^12 + T + 2
sage: phi.j_invariant(param[2])
T^35 + 2*T^31 + T^27 + 2*T^8 + T^4 + 2
```

```
>>> from sage.all import *
>>> A = GF(Integer(3))['T']
>>> K = Frac(A, names=('T',)); (T,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [T, T**Integer(2) + T + Integer(1), Integer(0), ↪
↪T**Integer(4) + Integer(1), T - Integer(1)])
>>> param = phi.basic_j_invariant_parameters(nonzero=True)
>>> phi.j_invariant(param[Integer(1)])
T^13 + 2*T^12 + T + 2
>>> phi.j_invariant(param[Integer(2)])
T^35 + 2*T^31 + T^27 + 2*T^8 + T^4 + 2
```

`jk_invariants()`

Return a dictionary whose keys are all the integers $1 \leq k \leq r - 1$ and the values are the corresponding j_k -invariants

Recall that the j_k -invariant of self is defined by:

$$j_k := \frac{g_k^{(q^r-1)/(\gcd(k,r)-1)}}{g_r^{(q^k-1)/(\gcd(k,r)-1)}}$$

where g_i is the i -th coefficient of the generator of self.

EXAMPLES:

```
sage: A = GF(3)['T']
sage: K.<T> = Frac(A)
sage: phi = DrinfeldModule(A, [T, 1, T+1, T^3, T^6])
sage: jk_inv = phi.jk_invariants(); jk_inv
{1: 1/T^6, 2: (T^10 + T^9 + T + 1)/T^6, 3: T^42}
sage: jk_inv[2]
(T^10 + T^9 + T + 1)/T^6
```

```
>>> from sage.all import *
>>> A = GF(Integer(3))['T']
>>> K = Frac(A, names=('T',)); (T,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [T, Integer(1), T+Integer(1), T**Integer(3), ↪
↪T**Integer(6)])
>>> jk_inv = phi.jk_invariants(); jk_inv
{1: 1/T^6, 2: (T^10 + T^9 + T + 1)/T^6, 3: T^42}
>>> jk_inv[Integer(2)]
(T^10 + T^9 + T + 1)/T^6
```

```

sage: F = GF(7**2)
sage: A = F['T']
sage: E.<z> = F.extension(4)
sage: phi = DrinfeldModule(A, [z^2, 1, z+1, z^2, z, z+1])
sage: phi.jk_invariants()
{1: 5*z^7 + 2*z^6 + 5*z^5 + 2*z^4 + 5*z^3 + z^2 + z + 2,
 2: 3*z^7 + 4*z^6 + 5*z^5 + 6*z^4 + 4*z,
 3: 5*z^7 + 6*z^6 + 6*z^5 + 4*z^3 + z^2 + 2*z + 1,
 4: 3*z^6 + 2*z^5 + 4*z^4 + 2*z^3 + 4*z^2 + 6*z + 2}

```

```

>>> from sage.all import *
>>> F = GF(Integer(7)**Integer(2))
>>> A = F['T']
>>> E = F.extension(Integer(4), names=('z',)); (z,) = E._first_ngens(1)
>>> phi = DrinfeldModule(A, [z**Integer(2), Integer(1), z+Integer(1),
↳ z**Integer(2), z, z+Integer(1)])
>>> phi.jk_invariants()
{1: 5*z^7 + 2*z^6 + 5*z^5 + 2*z^4 + 5*z^3 + z^2 + z + 2,
 2: 3*z^7 + 4*z^6 + 5*z^5 + 6*z^4 + 4*z,
 3: 5*z^7 + 6*z^6 + 6*z^5 + 4*z^3 + z^2 + 2*z + 1,
 4: 3*z^6 + 2*z^5 + 4*z^4 + 2*z^3 + 4*z^2 + 6*z + 2}

```

morphism()

Return the morphism object that defines the Drinfeld module.

OUTPUT: a ring morphism from the function ring to the Ore polynomial ring

EXAMPLES:

```

sage: Fq = GF(25)
sage: A.<T> = Fq[]
sage: K.<z12> = Fq.extension(6)
sage: p_root = 2*z12^11 + 2*z12^10 + z12^9 + 3*z12^8 + z12^7 + 2*z12^5 +
↳ 2*z12^4 + 3*z12^3 + z12^2 + 2*z12
sage: phi = DrinfeldModule(A, [p_root, z12^3, z12^5])
sage: phi.morphism()
Ring morphism:
  From: Univariate Polynomial Ring in T over Finite Field in z2 of size 5^2
  To:   Ore Polynomial Ring in t over Finite Field in z12 of size 5^12
        over its base twisted by Frob^2
  Defn: T |--> z12^5*t^2 + z12^3*t + 2*z12^11 + 2*z12^10 + z12^9 + 3*z12^8
          + z12^7 + 2*z12^5 + 2*z12^4 + 3*z12^3 + z12^2 + 2*z12
sage: from sage.rings.morphism import RingHomomorphism
sage: isinstance(phi.morphism(), RingHomomorphism)
True

```

```

>>> from sage.all import *
>>> Fq = GF(Integer(25))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(6), names=('z12',)); (z12,) = K._first_ngens(1)
>>> p_root = Integer(2)*z12**Integer(11) + Integer(2)*z12**Integer(10) +
↳ z12**Integer(9) + Integer(3)*z12**Integer(8) + z12**Integer(7) +
↳ Integer(2)*z12**Integer(5) + Integer(2)*z12**Integer(4) +
↳ Integer(3)*z12**Integer(3) + z12**Integer(2) + Integer(2)*z12
>>> phi = DrinfeldModule(A, [p_root, z12**Integer(3), z12**Integer(5)])
>>> phi.morphism()
Ring morphism:

```

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```

From: Univariate Polynomial Ring in T over Finite Field in z2 of size 5^2
To: Ore Polynomial Ring in t over Finite Field in z12 of size 5^12
    over its base twisted by Frob^2
Defn: T |--> z12^5*t^2 + z12^3*t + 2*z12^11 + 2*z12^10 + z12^9 + 3*z12^8
          + z12^7 + 2*z12^5 + 2*z12^4 + 3*z12^3 + z12^2 + 2*z12
>>> from sage.rings.morphism import RingHomomorphism
>>> isinstance(phi.morphism(), RingHomomorphism)
True

```

Actually, the DrinfeldModule method `__call__()` simply class the `__call__` method of this morphism:

```

sage: phi.morphism()(T) == phi(T)
True
sage: a = A.random_element()
sage: phi.morphism()(a) == phi(a)
True

```

```

>>> from sage.all import *
>>> phi.morphism()(T) == phi(T)
True
>>> a = A.random_element()
>>> phi.morphism()(a) == phi(a)
True

```

And many methods of the Drinfeld module have a counterpart in the morphism object:

```

sage: m = phi.morphism()
sage: m.domain() is phi.function_ring()
True
sage: m.codomain() is phi.ore_polring()
True
sage: m.im_gens()
[z12^5*t^2 + z12^3*t + 2*z12^11 + 2*z12^10 + z12^9 + 3*z12^8
 + z12^7 + 2*z12^5 + 2*z12^4 + 3*z12^3 + z12^2 + 2*z12]
sage: phi(T) == m.im_gens()[0]
True

```

```

>>> from sage.all import *
>>> m = phi.morphism()
>>> m.domain() is phi.function_ring()
True
>>> m.codomain() is phi.ore_polring()
True
>>> m.im_gens()
[z12^5*t^2 + z12^3*t + 2*z12^11 + 2*z12^10 + z12^9 + 3*z12^8
 + z12^7 + 2*z12^5 + 2*z12^4 + 3*z12^3 + z12^2 + 2*z12]
>>> phi(T) == m.im_gens()[Integer(0)]
True

```

rank()

Return the rank of the Drinfeld module.

In our case, the rank is the degree of the generator.

OUTPUT: an integer

EXAMPLES:

```

sage: Fq = GF(25)
sage: A.<T> = Fq[]
sage: K.<z12> = Fq.extension(6)
sage: p_root = 2*z12^11 + 2*z12^10 + z12^9 + 3*z12^8 + z12^7 + 2*z12^5 +
↪ 2*z12^4 + 3*z12^3 + z12^2 + 2*z12
sage: phi = DrinfeldModule(A, [p_root, z12^3, z12^5])
sage: phi.rank()
2
sage: psi = DrinfeldModule(A, [p_root, 2])
sage: psi.rank()
1
sage: rho = DrinfeldModule(A, [p_root, 0, 0, 0, 1])
sage: rho.rank()
4

```

```

>>> from sage.all import *
>>> Fq = GF(Integer(25))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(6), names=('z12',)); (z12,) = K._first_ngens(1)
>>> p_root = Integer(2)*z12**Integer(11) + Integer(2)*z12**Integer(10) +
↪ z12**Integer(9) + Integer(3)*z12**Integer(8) + z12**Integer(7) +
↪ Integer(2)*z12**Integer(5) + Integer(2)*z12**Integer(4) +
↪ Integer(3)*z12**Integer(3) + z12**Integer(2) + Integer(2)*z12
>>> phi = DrinfeldModule(A, [p_root, z12**Integer(3), z12**Integer(5)])
>>> phi.rank()
2
>>> psi = DrinfeldModule(A, [p_root, Integer(2)])
>>> psi.rank()
1
>>> rho = DrinfeldModule(A, [p_root, Integer(0), Integer(0), Integer(0),
↪ Integer(1)])
>>> rho.rank()
4

```

scalar_multiplication (x)

Return the endomorphism of this Drinfeld module, which is the multiplication by x , i.e. the isogeny defined by the Ore polynomial ϕ_x .

INPUT:

- x – an element in the ring of functions

EXAMPLES:

```

sage: Fq = GF(5)
sage: A.<T> = Fq[]
sage: K.<z> = Fq.extension(3)
sage: phi = DrinfeldModule(A, [z, 0, 1, z])
sage: phi
Drinfeld module defined by T |--> z*t^3 + t^2 + z
sage: phi.hom(T) # indirect doctest
Endomorphism of Drinfeld module defined by T |--> z*t^3 + t^2 + z
Defn: z*t^3 + t^2 + z

```

```

>>> from sage.all import *
>>> Fq = GF(Integer(5))

```

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```

>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(3), names=('z',)); (z,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [z, Integer(0), Integer(1), z])
>>> phi
Drinfeld module defined by T |--> z*t^3 + t^2 + z
>>> phi.hom(T) # indirect doctest
Endomorphism of Drinfeld module defined by T |--> z*t^3 + t^2 + z
Defn: z*t^3 + t^2 + z

```

```

sage: phi.hom(T^2 + 1)
Endomorphism of Drinfeld module defined by T |--> z*t^3 + t^2 + z
Defn: z^2*t^6 + (3*z^2 + z + 1)*t^5 + t^4 + 2*z^2*t^3 + (3*z^2 + z + 1)*t^2 +
+ z^2 + 1

```

```

>>> from sage.all import *
>>> phi.hom(T**Integer(2) + Integer(1))
Endomorphism of Drinfeld module defined by T |--> z*t^3 + t^2 + z
Defn: z^2*t^6 + (3*z^2 + z + 1)*t^5 + t^4 + 2*z^2*t^3 + (3*z^2 + z + 1)*t^2 +
+ z^2 + 1

```

velu (*isog*)

Return a new Drinfeld module such that input is an isogeny to this module with domain `self`; if no such isogeny exists, raise an exception.

INPUT:

- `isog` – the Ore polynomial that defines the isogeny

OUTPUT: a Drinfeld module

ALGORITHM:

The input defines an isogeny if only if:

1. The degree of the characteristic divides the height of the input. (The height of an Ore polynomial $P(\tau)$ is the maximum n such that τ^n right-divides $P(\tau)$.)
2. The input right-divides the generator, which can be tested with Euclidean division.

We test if the input is an isogeny, and, if it is, we return the quotient of the Euclidean division.

Height and Euclidean division of Ore polynomials are implemented as methods of class `sage.rings.polynomial.ore_polynomial_element.OrePolynomial`.

Another possible algorithm is to recursively solve a system, see [arXiv 2203.06970](#), Eq. 1.1.

EXAMPLES:

```

sage: Fq = GF(25)
sage: A.<T> = Fq[]
sage: K.<z12> = Fq.extension(6)
sage: p_root = 2*z12^11 + 2*z12^10 + z12^9 + 3*z12^8 + z12^7 + 2*z12^5 +
+ 2*z12^4 + 3*z12^3 + z12^2 + 2*z12
sage: phi = DrinfeldModule(A, [p_root, z12^3, z12^5])
sage: t = phi.ore_polring().gen()
sage: isog = t + 2*z12^11 + 4*z12^9 + 2*z12^8 + 2*z12^6 + 3*z12^5 + z12^4 +
+ 2*z12^3 + 4*z12^2 + 4*z12 + 4
sage: psi = phi.velu(isog)
sage: psi

```

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```

Drinfeld module defined by T |-->
(z12^11 + 3*z12^10 + z12^9 + z12^7 + z12^5 + 4*z12^4 + 4*z12^3 + z12^2 +
↪1)*t^2
+ (2*z12^11 + 4*z12^10 + 2*z12^8 + z12^6 + 3*z12^5 + z12^4 + 2*z12^3 + z12^2
↪+ z12 + 4)*t
+ 2*z12^11 + 2*z12^10 + z12^9 + 3*z12^8 + z12^7 + 2*z12^5 + 2*z12^4 + 3*z12^
↪3 + z12^2 + 2*z12
sage: isog in Hom(phi, psi)
True

```

```

>>> from sage.all import *
>>> Fq = GF(Integer(25))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(6), names=('z12',)); (z12,) = K._first_ngens(1)
>>> p_root = Integer(2)*z12**Integer(11) + Integer(2)*z12**Integer(10) +
↪z12**Integer(9) + Integer(3)*z12**Integer(8) + z12**Integer(7) +
↪Integer(2)*z12**Integer(5) + Integer(2)*z12**Integer(4) +
↪Integer(3)*z12**Integer(3) + z12**Integer(2) + Integer(2)*z12
>>> phi = DrinfeldModule(A, [p_root, z12**Integer(3), z12**Integer(5)])
>>> t = phi.ore_polring().gen()
>>> isog = t + Integer(2)*z12**Integer(11) + Integer(4)*z12**Integer(9) +
↪Integer(2)*z12**Integer(8) + Integer(2)*z12**Integer(6) +
↪Integer(3)*z12**Integer(5) + z12**Integer(4) + Integer(2)*z12**Integer(3) +
↪Integer(4)*z12**Integer(2) + Integer(4)*z12 + Integer(4)
>>> psi = phi.velu(isog)
>>> psi
Drinfeld module defined by T |-->
(z12^11 + 3*z12^10 + z12^9 + z12^7 + z12^5 + 4*z12^4 + 4*z12^3 + z12^2 +
↪1)*t^2
+ (2*z12^11 + 4*z12^10 + 2*z12^8 + z12^6 + 3*z12^5 + z12^4 + 2*z12^3 + z12^2
↪+ z12 + 4)*t
+ 2*z12^11 + 2*z12^10 + z12^9 + 3*z12^8 + z12^7 + 2*z12^5 + 2*z12^4 + 3*z12^
↪3 + z12^2 + 2*z12
>>> isog in Hom(phi, psi)
True

```

This method works for endomorphisms as well:

```

sage: phi.velu(phi(T)) is phi
True
sage: phi.velu(t^6) is phi
True

```

```

>>> from sage.all import *
>>> phi.velu(phi(T)) is phi
True
>>> phi.velu(t**Integer(6)) is phi
True

```

The following inputs do not define isogenies, and the method returns None:

```

sage: phi.velu(0)
Traceback (most recent call last):
...
ValueError: the input does not define an isogeny
sage: phi.velu(t)

```

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```
Traceback (most recent call last):
...
ValueError: the input does not define an isogeny
sage: phi.velu(t^3 + t + 2)
Traceback (most recent call last):
...
ValueError: the input does not define an isogeny
```

```
>>> from sage.all import *
>>> phi.velu(Integer(0))
Traceback (most recent call last):
...
ValueError: the input does not define an isogeny
>>> phi.velu(t)
Traceback (most recent call last):
...
ValueError: the input does not define an isogeny
>>> phi.velu(t**Integer(3) + t + Integer(2))
Traceback (most recent call last):
...
ValueError: the input does not define an isogeny
```

1.2 Drinfeld modules over rings of characteristic zero

This module provides the class `sage.rings.function_fields.drinfeld_module.charzero_drinfeld_module.DrinfeldModule_charzero`, which inherits `sage.rings.function_fields.drinfeld_module.drinfeld_module.DrinfeldModule`.

AUTHORS:

- David Ayotte (2023-09)

```
class sage.rings.function_field.drinfeld_modules.charzero_drinfeld_module.DrinfeldModule_cl
```

Bases: *DrinfeldModule*

This class implements Drinfeld $\mathbb{F}_q[T]$ -modules defined over fields of $\mathbb{F}_q[T]$ -characteristic zero.

Recall that the $\mathbb{F}_q[T]$ -characteristic is defined as the kernel of the underlying structure morphism. For general definitions and help on Drinfeld modules, see class `sage.rings.function_fields.drinfeld_module.drinfeld_module.DrinfeldModule`.

Construction:

The user does not ever need to directly call `DrinfeldModule_charzero` — the metaclass `DrinfeldModule` is responsible for instantiating the right class depending on the input:

```
sage: A = GF(3)['T']
sage: K.<T> = Frac(A)
sage: phi = DrinfeldModule(A, [T, 1])
sage: phi
Drinfeld module defined by T |--> t + T
```

```
>>> from sage.all import *
>>> A = GF(Integer(3))['T']
>>> K = Frac(A, names=('T',)); (T,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [T, Integer(1)])
>>> phi
Drinfeld module defined by T |--> t + T
```

```
sage: isinstance(phi, DrinfeldModule)
True
sage: from sage.rings.function_field.drinfeld_modules.charzero_drinfeld_module_
↳import DrinfeldModule_charzero
sage: isinstance(phi, DrinfeldModule_charzero)
True
```

```
>>> from sage.all import *
>>> isinstance(phi, DrinfeldModule)
True
>>> from sage.rings.function_field.drinfeld_modules.charzero_drinfeld_module_
↳import DrinfeldModule_charzero
>>> isinstance(phi, DrinfeldModule_charzero)
True
```

Logarithm and exponential

It is possible to calculate the logarithm and the exponential of any Drinfeld modules of characteristic zero:

```
sage: A = GF(2)['T']
sage: K.<T> = Frac(A)
sage: phi = DrinfeldModule(A, [T, 1])
sage: phi.exponential()
z + ((1/(T^2+T))*z^2) + ((1/(T^8+T^6+T^5+T^3))*z^4) + O(z^8)
sage: phi.logarithm()
z + ((1/(T^2+T))*z^2) + ((1/(T^6+T^5+T^3+T^2))*z^4) + O(z^8)
```

```
>>> from sage.all import *
>>> A = GF(Integer(2))['T']
>>> K = Frac(A, names=('T',)); (T,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [T, Integer(1)])
>>> phi.exponential()
z + ((1/(T^2+T))*z^2) + ((1/(T^8+T^6+T^5+T^3))*z^4) + O(z^8)
>>> phi.logarithm()
z + ((1/(T^2+T))*z^2) + ((1/(T^6+T^5+T^3+T^2))*z^4) + O(z^8)
```

Goss polynomials

Goss polynomials are a sequence of polynomials related with the analytic theory of Drinfeld module. They provide a function field analogue of certain classical trigonometric functions:

```
sage: A = GF(2) ['T']
sage: K.<T> = Frac(A)
sage: phi = DrinfeldModule(A, [T, 1])
sage: phi.goss_polynomial(1)
X
sage: phi.goss_polynomial(2)
X^2
sage: phi.goss_polynomial(3)
X^3 + (1/(T^2 + T))*X^2
```

```
>>> from sage.all import *
>>> A = GF(Integer(2)) ['T']
>>> K = Frac(A, names=('T',)); (T,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [T, Integer(1)])
>>> phi.goss_polynomial(Integer(1))
X
>>> phi.goss_polynomial(Integer(2))
X^2
>>> phi.goss_polynomial(Integer(3))
X^3 + (1/(T^2 + T))*X^2
```

Base fields of $\mathbb{F}_q[T]$ -characteristic zero

The base fields need not only be fraction fields of polynomials ring. In the following example, we construct a Drinfeld module over $\mathbb{F}_q((1/T))$, the completion of the rational function field at the place $1/T$:

```
sage: A.<T> = GF(2) []
sage: L.<s> = LaurentSeriesRing(GF(2)) # s = 1/T
sage: phi = DrinfeldModule(A, [1/s, s + s^2 + s^5 + O(s^6), 1+1/s])
sage: phi(T)
(s^-1 + 1)*t^2 + (s + s^2 + s^5 + O(s^6))*t + s^-1
```

```
>>> from sage.all import *
>>> A = GF(Integer(2)) ['T']; (T,) = A._first_ngens(1)
>>> L = LaurentSeriesRing(GF(Integer(2)), names=('s',)); (s,) = L._first_ngens(1)
↪ # s = 1/T
>>> phi = DrinfeldModule(A, [Integer(1)/s, s + s**Integer(2) + s**Integer(5) +
↪ O(s**Integer(6)), Integer(1)+Integer(1)/s])
>>> phi(T)
(s^-1 + 1)*t^2 + (s + s^2 + s^5 + O(s^6))*t + s^-1
```

One can also construct Drinfeld modules over SageMath's global function fields:

```
sage: A.<T> = GF(5) []
sage: K.<z> = FunctionField(GF(5)) # z = T
sage: phi = DrinfeldModule(A, [z, 1, z^2])
sage: phi(T)
z^2*t^2 + t + z
```

```

>>> from sage.all import *
>>> A = GF(Integer(5))['T']; (T,) = A._first_ngens(1)
>>> K = FunctionField(GF(Integer(5)), names=('z',)); (z,) = K._first_ngens(1) # z ↦ T
>>> phi = DrinfeldModule(A, [z, Integer(1), z**Integer(2)])
>>> phi(T)
z^2*t^2 + t + z

```

exponential (name='z')

Return the exponential of this Drinfeld module.

Note that the exponential is only defined when the $\mathbb{F}_q[T]$ -characteristic is zero.

INPUT:

- name (string, default: 'z') – the name of the generator of the lazy power series ring.

OUTPUT:

A lazy power series over the base field.

EXAMPLES:

```

sage: A = GF(2)['T']
sage: K.<T> = Frac(A)
sage: phi = DrinfeldModule(A, [T, 1])
sage: q = A.base_ring().cardinality()
sage: exp = phi.exponential(); exp
z + ((1/(T^2+T))*z^2) + ((1/(T^8+T^6+T^5+T^3))*z^4) + O(z^8)

```

```

>>> from sage.all import *
>>> A = GF(Integer(2))['T']
>>> K = Frac(A, names=('T',)); (T,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [T, Integer(1)])
>>> q = A.base_ring().cardinality()
>>> exp = phi.exponential(); exp
z + ((1/(T^2+T))*z^2) + ((1/(T^8+T^6+T^5+T^3))*z^4) + O(z^8)

```

The exponential is returned as a lazy power series, meaning that any of its coefficients can be computed on demands:

```

sage: exp[2^4]
1/(T^64 + T^56 + T^52 + ... + T^27 + T^23 + T^15)
sage: exp[2^5]
1/(T^160 + T^144 + T^136 + ... + T^55 + T^47 + T^31)

```

```

>>> from sage.all import *
>>> exp[Integer(2)**Integer(4)]
1/(T^64 + T^56 + T^52 + ... + T^27 + T^23 + T^15)
>>> exp[Integer(2)**Integer(5)]
1/(T^160 + T^144 + T^136 + ... + T^55 + T^47 + T^31)

```

Example in higher rank:

```

sage: A = GF(5)['T']
sage: K.<T> = Frac(A)
sage: phi = DrinfeldModule(A, [T, T^2, T + T^2 + T^4, 1])
sage: exp = phi.exponential(); exp
z + ((T/(T^4+4))*z^5) + O(z^8)

```

```
>>> from sage.all import *
>>> A = GF(Integer(5))['T']
>>> K = Frac(A, names=('T',)); (T,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [T, T**Integer(2), T + T**Integer(2) +
↪ T**Integer(4), Integer(1)])
>>> exp = phi.exponential(); exp
z + ((T/(T^4+4))*z^5) + O(z^8)
```

The exponential is the compositional inverse of the logarithm (see `logarithm()`):

```
sage: log = phi.logarithm(); log
z + ((4*T/(T^4+4))*z^5) + O(z^8)
sage: exp.compose(log)
z + O(z^8)
sage: log.compose(exp)
z + O(z^8)
```

```
>>> from sage.all import *
>>> log = phi.logarithm(); log
z + ((4*T/(T^4+4))*z^5) + O(z^8)
>>> exp.compose(log)
z + O(z^8)
>>> log.compose(exp)
z + O(z^8)
```

REFERENCE:

See section 4.6 of [Gos1998] for the definition of the exponential.

goss_polynomial (*n*, *var*='X')

Return the *n*-th Goss polynomial of the Drinfeld module.

Note that Goss polynomials are only defined for Drinfeld modules of characteristic zero.

INPUT:

- *n* (integer) – the index of the Goss polynomial
- *var* (str, default: 'X') – the name of polynomial variable.

OUTPUT:

- a univariate polynomial in *var* over the base *A*-field.

EXAMPLES:

```
sage: A = GF(3)['T']
sage: K.<T> = Frac(A)
sage: phi = DrinfeldModule(A, [T, 1]) # Carlitz module
sage: phi.goss_polynomial(1)
X
sage: phi.goss_polynomial(2)
X^2
sage: phi.goss_polynomial(4)
X^4 + (1/(T^3 + 2*T))*X^2
sage: phi.goss_polynomial(5)
X^5 + (2/(T^3 + 2*T))*X^3
sage: phi.goss_polynomial(10)
X^10 + (1/(T^3 + 2*T))*X^8 + (1/(T^6 + T^4 + T^2))*X^6 + (1/(T^9 + 2*T^3))*X^
↪ 4 + (1/(T^18 + 2*T^12 + 2*T^10 + T^4))*X^2
```

```

>>> from sage.all import *
>>> A = GF(Integer(3))['T']
>>> K = Frac(A, names=('T',)); (T,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [T, Integer(1)]) # Carlitz module
>>> phi.goss_polynomial(Integer(1))
X
>>> phi.goss_polynomial(Integer(2))
X^2
>>> phi.goss_polynomial(Integer(4))
X^4 + (1/(T^3 + 2*T))*X^2
>>> phi.goss_polynomial(Integer(5))
X^5 + (2/(T^3 + 2*T))*X^3
>>> phi.goss_polynomial(Integer(10))
X^10 + (1/(T^3 + 2*T))*X^8 + (1/(T^6 + T^4 + T^2))*X^6 + (1/(T^9 + 2*T^3))*X^
↪ 4 + (1/(T^18 + 2*T^12 + 2*T^10 + T^4))*X^2

```

REFERENCE:

Section 3 of [Gek1988] provides an exposition of Goss polynomials.

logarithm (name='z')

Return the logarithm of the given Drinfeld module.

By definition, the logarithm is the compositional inverse of the exponential (see [exponential\(\)](#)). Note that the logarithm is only defined when the $\mathbb{F}_q[T]$ -characteristic is zero.

INPUT:

- name (string, default: 'z') – the name of the generator of the lazy power series ring.

OUTPUT:

A lazy power series over the base field.

EXAMPLES:

```

sage: A = GF(2)['T']
sage: K.<T> = Frac(A)
sage: phi = DrinfeldModule(A, [T, 1])
sage: log = phi.logarithm(); log
z + ((1/(T^2+T))*z^2) + ((1/(T^6+T^5+T^3+T^2))*z^4) + O(z^8)

```

```

>>> from sage.all import *
>>> A = GF(Integer(2))['T']
>>> K = Frac(A, names=('T',)); (T,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [T, Integer(1)])
>>> log = phi.logarithm(); log
z + ((1/(T^2+T))*z^2) + ((1/(T^6+T^5+T^3+T^2))*z^4) + O(z^8)

```

The logarithm is returned as a lazy power series, meaning that any of its coefficients can be computed on demands:

```

sage: log[2^4]
1/(T^30 + T^29 + T^27 + ... + T^7 + T^5 + T^4)
sage: log[2^5]
1/(T^62 + T^61 + T^59 + ... + T^8 + T^6 + T^5)

```

```

>>> from sage.all import *
>>> log[Integer(2)**Integer(4)]

```

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```

1/(T^30 + T^29 + T^27 + ... + T^7 + T^5 + T^4)
>>> log(Integer(2)**Integer(5))
1/(T^62 + T^61 + T^59 + ... + T^8 + T^6 + T^5)

```

Example in higher rank:

```

sage: A = GF(5)['T']
sage: K.<T> = Frac(A)
sage: phi = DrinfeldModule(A, [T, T^2, T + T^2 + T^4, 1])
sage: phi.logarithm()
z + ((4*T/(T^4+4))*z^5) + O(z^8)

```

```

>>> from sage.all import *
>>> A = GF(Integer(5))['T']
>>> K = Frac(A, names=('T',)); (T,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [T, T**Integer(2), T + T**Integer(2) +
↳ T**Integer(4), Integer(1)])
>>> phi.logarithm()
z + ((4*T/(T^4+4))*z^5) + O(z^8)

```

1.3 Finite Drinfeld modules

This module provides the class `sage.rings.function_fields.drinfeld_module.finite_drinfeld_module.DrinfeldModule_finite`, which inherits `sage.rings.function_fields.drinfeld_module.drinfeld_module.DrinfeldModule`.

AUTHORS:

- Antoine Leudière (2022-04)
- Yossef Musleh (2023-02): added characteristic polynomial methods

class `sage.rings.function_field.drinfeld_modules.finite_drinfeld_module.DrinfeldModule_finite`

Bases: `DrinfeldModule`

This class implements finite Drinfeld $\mathbb{F}_q[T]$ -modules.

A *finite Drinfeld module* is a Drinfeld module whose base field is finite. In this case, the function field characteristic is a prime ideal.

For general definitions and help on Drinfeld modules, see class `sage.rings.function_fields.drinfeld_module.drinfeld_module.DrinfeldModule`.

Construction:

The user does not ever need to directly call `DrinfeldModule_finite` — the metaclass `DrinfeldModule` is responsible for instantiating `DrinfeldModule` or `DrinfeldModule_finite` depending on the input:

```
sage: Fq = GF(343)
sage: A.<T> = Fq[]
sage: K.<z6> = Fq.extension(2)
sage: phi = DrinfeldModule(A, [z6, 0, 5])
sage: phi
Drinfeld module defined by T |--> 5*t^2 + z6
```

```
>>> from sage.all import *
>>> Fq = GF(Integer(343))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(2), names=('z6',)); (z6,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [z6, Integer(0), Integer(5)])
>>> phi
Drinfeld module defined by T |--> 5*t^2 + z6
```

```
sage: isinstance(phi, DrinfeldModule)
True
sage: from sage.rings.function_field.drinfeld_modules.finite_drinfeld_module_
↳import DrinfeldModule_finite
sage: isinstance(phi, DrinfeldModule_finite)
True
```

```
>>> from sage.all import *
>>> isinstance(phi, DrinfeldModule)
True
>>> from sage.rings.function_field.drinfeld_modules.finite_drinfeld_module import_
↳DrinfeldModule_finite
>>> isinstance(phi, DrinfeldModule_finite)
True
```

The user should never use `DrinfeldModule_finite` to test if a Drinfeld module is finite, but rather the `is_finite` method:

```
sage: phi.is_finite()
True
```

```
>>> from sage.all import *
>>> phi.is_finite()
True
```

Complex multiplication of rank two finite Drinfeld modules

We can handle some aspects of the theory of complex multiplication of finite Drinfeld modules. Apart from the method `frobenius_endomorphism`, we only handle rank two Drinfeld modules.

First of all, it is easy to create the Frobenius endomorphism:

```
sage: frobenius_endomorphism = phi.frobenius_endomorphism()
sage: frobenius_endomorphism
```

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```
Endomorphism of Drinfeld module defined by T |--> 5*t^2 + z6
Defn: t^2
```

```
>>> from sage.all import *
>>> frobenius_endomorphism = phi.frobenius_endomorphism()
>>> frobenius_endomorphism
Endomorphism of Drinfeld module defined by T |--> 5*t^2 + z6
Defn: t^2
```

Its characteristic polynomial can be computed:

```
sage: chi = phi.frobenius_charpoly()
sage: chi
X^2 + (T + 2*z3^2 + 2*z3 + 1)*X + 2*T^2 + (z3^2 + z3 + 4)*T + 2*z3
sage: frob_pol = frobenius_endomorphism.ore_polynomial()
sage: chi(frob_pol, phi(T))
0
```

```
>>> from sage.all import *
>>> chi = phi.frobenius_charpoly()
>>> chi
X^2 + (T + 2*z3^2 + 2*z3 + 1)*X + 2*T^2 + (z3^2 + z3 + 4)*T + 2*z3
>>> frob_pol = frobenius_endomorphism.ore_polynomial()
>>> chi(frob_pol, phi(T))
0
```

as well as its trace and norm:

```
sage: phi.frobenius_trace()
6*T + 5*z3^2 + 5*z3 + 6
sage: phi.frobenius_trace() == -chi[1]
True
sage: phi.frobenius_norm()
2*T^2 + (z3^2 + z3 + 4)*T + 2*z3
```

```
>>> from sage.all import *
>>> phi.frobenius_trace()
6*T + 5*z3^2 + 5*z3 + 6
>>> phi.frobenius_trace() == -chi[Integer(1)]
True
>>> phi.frobenius_norm()
2*T^2 + (z3^2 + z3 + 4)*T + 2*z3
```

We can decide if a Drinfeld module is ordinary or supersingular:

```
sage: phi.is_ordinary()
True
sage: phi.is_supersingular()
False
```

```
>>> from sage.all import *
>>> phi.is_ordinary()
True
>>> phi.is_supersingular()
False
```

Inverting the Drinfeld module

The morphism that defines a Drinfeld module is injective (see [Gos1998], cor. 4.5.2). If the Drinfeld module is finite, one can retrieve preimages:

```
sage: a = A.random_element()
sage: phi.invert(phi(a)) == a
True
```

```
>>> from sage.all import *
>>> a = A.random_element()
>>> phi.invert(phi(a)) == a
True
```

frobenius_charpoly (var='X', algorithm='crystalline')

Return the characteristic polynomial of the Frobenius endomorphism.

Let $\mathbb{F}_q$ be the base field of the function ring. The *characteristic polynomial* χ of the Frobenius endomorphism is defined in [Gek1991]. An important feature of this polynomial is that it is monic, univariate, and has coefficients in the function ring. As in our case the function ring is a univariate polynomial ring, it is customary to see the characteristic polynomial of the Frobenius endomorphism as a bivariate polynomial.

Let $\chi = X^r + \sum_{i=0}^{r-1} A_i(T)X^i$ be the characteristic polynomial of the Frobenius endomorphism, and let t^n be the Ore polynomial that defines the Frobenius endomorphism of ϕ ; by definition, n is the degree of K over the base field $\mathbb{F}_q$. Then we have

$$\chi(t^n)(\phi(T)) = t^{nr} + \sum_{i=1}^r \phi_{A_i} t^{n(i)} = 0,$$

with $\deg(A_i) \leq \frac{n(r-i)}{r}$.

Note that the *Frobenius trace* is defined as $A_{r-1}(T)$ and the *Frobenius norm* is defined as $A_0(T)$.

INPUT:

- var (default: 'X') – the name of the second variable
- algorithm (default: 'crystalline') – the algorithm used to compute the characteristic polynomial

EXAMPLES:

```
sage: Fq = GF(25)
sage: A.<T> = Fq[]
sage: K.<z12> = Fq.extension(6)
sage: p_root = 2*z12^11 + 2*z12^10 + z12^9 + 3*z12^8 + z12^7 + 2*z12^5 +
↪ 2*z12^4 + 3*z12^3 + z12^2 + 2*z12
sage: phi = DrinfeldModule(A, [p_root, z12^3, z12^5])
sage: phi.frobenius_charpoly()
X^2 + ((4*z2 + 4)*T^3 + (z2 + 3)*T^2 + 3*T + 2*z2 + 3)*X + 3*z2*T^6 + (4*z2 +
↪ 3)*T^5 + (4*z2 + 4)*T^4 + 2*T^3 + (3*z2 + 3)*T^2 + (z2 + 2)*T + 4*z2
```

```
>>> from sage.all import *
>>> Fq = GF(Integer(25))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(6), names=('z12',)); (z12,) = K._first_ngens(1)
>>> p_root = Integer(2)*z12**Integer(11) + Integer(2)*z12**Integer(10) +
↪ z12**Integer(9) + Integer(3)*z12**Integer(8) + z12**Integer(7) +
```

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```

↪Integer(2)*z12**Integer(5) + Integer(2)*z12**Integer(4) +_
↪Integer(3)*z12**Integer(3) + z12**Integer(2) + Integer(2)*z12
>>> phi = DrinfeldModule(A, [p_root, z12**Integer(3), z12**Integer(5)])
>>> phi.frobenius_charpoly()
X^2 + ((4*z2 + 4)*T^3 + (z2 + 3)*T^2 + 3*T + 2*z2 + 3)*X + 3*z2*T^6 + (4*z2 +_
↪3)*T^5 + (4*z2 + 4)*T^4 + 2*T^3 + (3*z2 + 3)*T^2 + (z2 + 2)*T + 4*z2

```

```

sage: Fq = GF(343)
sage: A.<T> = Fq[]
sage: K.<z6> = Fq.extension(2)
sage: phi = DrinfeldModule(A, [1, 0, z6])
sage: chi = phi.frobenius_charpoly()
sage: chi
X^2 + ((3*z3^2 + z3 + 4)*T + 4*z3^2 + 6*z3 + 3)*X + (5*z3^2 + 2*z3)*T^2 +_
↪(4*z3^2 + 3*z3)*T + 5*z3^2 + 2*z3

```

```

>>> from sage.all import *
>>> Fq = GF(Integer(343))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(2), names=('z6',)); (z6,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [Integer(1), Integer(0), z6])
>>> chi = phi.frobenius_charpoly()
>>> chi
X^2 + ((3*z3^2 + z3 + 4)*T + 4*z3^2 + 6*z3 + 3)*X + (5*z3^2 + 2*z3)*T^2 +_
↪(4*z3^2 + 3*z3)*T + 5*z3^2 + 2*z3

```

```

sage: frob_pol = phi.frobenius_endomorphism().ore_polynomial()
sage: chi(frob_pol, phi(T))
0

```

```

>>> from sage.all import *
>>> frob_pol = phi.frobenius_endomorphism().ore_polynomial()
>>> chi(frob_pol, phi(T))
0

```

```

sage: phi.frobenius_charpoly(algorithm="NotImplemented")
Traceback (most recent call last):
...
NotImplementedError: algorithm "NotImplemented" not implemented

```

```

>>> from sage.all import *
>>> phi.frobenius_charpoly(algorithm="NotImplemented")
Traceback (most recent call last):
...
NotImplementedError: algorithm "NotImplemented" not implemented

```

ALGORITHM:

By default, this method uses the so-called *crystalline* algorithm which computes the characteristic polynomial of the Frobenius acting on the crystalline cohomology of the Drinfeld module. For further details, see [Ang1997].

The available options for ‘algorithm’ are:

- ‘crystalline’ – Computes the characteristic polynomial of the Frobenius endomorphism on the crystalline cohomology of a Drinfeld module.

- 'motive' – Based on computing the characteristic polynomial of the Frobenius endomorphism on the motive of a Drinfeld module. This instantiates the Frobenius as a morphism object and calls its 'characteristic_polynomial' method.

frobenius_endomorphism()

Return the Frobenius endomorphism of the Drinfeld module as a morphism object.

Let q be the order of the base field of the function ring. The *Frobenius endomorphism* is defined as the endomorphism whose defining Ore polynomial is t^q .

EXAMPLES:

```
sage: Fq = GF(343)
sage: A.<T> = Fq[]
sage: K.<z6> = Fq.extension(2)
sage: phi = DrinfeldModule(A, [1, 0, z6])
sage: phi.frobenius_endomorphism()
Endomorphism of Drinfeld module defined by T |--> z6*t^2 + 1
Defn: t^2
```

```
>>> from sage.all import *
>>> Fq = GF(Integer(343))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(2), names=('z6',)); (z6,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [Integer(1), Integer(0), z6])
>>> phi.frobenius_endomorphism()
Endomorphism of Drinfeld module defined by T |--> z6*t^2 + 1
Defn: t^2
```

frobenius_norm()

Return the Frobenius norm of the Drinfeld module.

Let $C(X) = \sum_{i=0}^r a_i X^i$ denote the characteristic polynomial of the Frobenius endomorphism. The Frobenius norm is $(-1)^r a_0$. This is an element of the regular function ring and if n is the degree of the base field over $\mathbb{F}_q$, then the Frobenius norm has degree n .

EXAMPLES:

```
sage: Fq = GF(343)
sage: A.<T> = Fq[]
sage: K.<z6> = Fq.extension(2)
sage: phi = DrinfeldModule(A, [1, 0, z6])
sage: B = phi.frobenius_norm()
sage: B
(5*z3^2 + 2*z3)*T^2 + (4*z3^2 + 3*z3)*T + 5*z3^2 + 2*z3
```

```
>>> from sage.all import *
>>> Fq = GF(Integer(343))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(2), names=('z6',)); (z6,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [Integer(1), Integer(0), z6])
>>> B = phi.frobenius_norm()
>>> B
(5*z3^2 + 2*z3)*T^2 + (4*z3^2 + 3*z3)*T + 5*z3^2 + 2*z3
```

```
sage: n = 2 # Degree of the base field over Fq
sage: B.degree() == n
True
```

```
>>> from sage.all import *
>>> n = Integer(2) # Degree of the base field over Fq
>>> B.degree() == n
True
```

```
sage: B == phi.frobenius_charpoly()[0]
True
```

```
>>> from sage.all import *
>>> B == phi.frobenius_charpoly()[Integer(0)]
True
```

ALGORITHM:

The Frobenius norm is computed using the formula, by Gekeler, given in [MS2019], Section 4, Proposition 3.

frobenius_trace()

Return the Frobenius trace of the Drinfeld module.

Let $C(X) = \sum_{i=0}^r a_i X^i$ denote the characteristic polynomial of the Frobenius endomorphism. The Frobenius trace is $-a_{r-1}$. This is an element of the regular function ring and if n is the degree of the base field over $\mathbb{F}_q$, then the Frobenius trace has degree at most $\frac{n}{r}$.

EXAMPLES:

```
sage: Fq = GF(343)
sage: A.<T> = Fq[]
sage: K.<z6> = Fq.extension(2)
sage: phi = DrinfeldModule(A, [1, 0, z6])
sage: A = phi.frobenius_trace()
sage: A
(4*z3^2 + 6*z3 + 3)*T + 3*z3^2 + z3 + 4
```

```
>>> from sage.all import *
>>> Fq = GF(Integer(343))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(2), names=('z6',)); (z6,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [Integer(1), Integer(0), z6])
>>> A = phi.frobenius_trace()
>>> A
(4*z3^2 + 6*z3 + 3)*T + 3*z3^2 + z3 + 4
```

```
sage: n = 2 # Degree over Fq of the base codomain
sage: A.degree() <= n/2
True
```

```
>>> from sage.all import *
>>> n = Integer(2) # Degree over Fq of the base codomain
>>> A.degree() <= n/Integer(2)
True
```

```
sage: A == -phi.frobenius_charpoly()[1]
True
```

```
>>> from sage.all import *
>>> A == -phi.frobenius_charpoly()[Integer(1)]
True
```

ALGORITHM:

We extract the coefficient of X^{r-1} from the characteristic polynomial if it has been previously computed, otherwise we compute the trace of the matrix of the Frobenius acting on the crystalline cohomology.

invert (*ore_pol*)

Return the preimage of the input under the Drinfeld module, if it exists.

INPUT:

- *ore_pol* – the Ore polynomial whose preimage we want to compute

EXAMPLES:

```
sage: Fq = GF(25)
sage: A.<T> = Fq[]
sage: K.<z12> = Fq.extension(6)
sage: p_root = 2*z12^11 + 2*z12^10 + z12^9 + 3*z12^8 + z12^7 + 2*z12^5 +
↳ 2*z12^4 + 3*z12^3 + z12^2 + 2*z12
sage: phi = DrinfeldModule(A, [p_root, z12^3, z12^5])
sage: a = A.random_element()
sage: phi.invert(phi(a)) == a
True
sage: phi.invert(phi(T)) == T
True
sage: phi.invert(phi(Fq.gen())) == Fq.gen()
True
```

```
>>> from sage.all import *
>>> Fq = GF(Integer(25))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(6), names=('z12',)); (z12,) = K._first_ngens(1)
>>> p_root = Integer(2)*z12**Integer(11) + Integer(2)*z12**Integer(10) +
↳ z12**Integer(9) + Integer(3)*z12**Integer(8) + z12**Integer(7) +
↳ Integer(2)*z12**Integer(5) + Integer(2)*z12**Integer(4) +
↳ Integer(3)*z12**Integer(3) + z12**Integer(2) + Integer(2)*z12
>>> phi = DrinfeldModule(A, [p_root, z12**Integer(3), z12**Integer(5)])
>>> a = A.random_element()
>>> phi.invert(phi(a)) == a
True
>>> phi.invert(phi(T)) == T
True
>>> phi.invert(phi(Fq.gen())) == Fq.gen()
True
```

When the input is not in the image of the Drinfeld module, an exception is raised:

```
sage: t = phi.ore_polring().gen()
sage: phi.invert(t + 1)
Traceback (most recent call last):
...
ValueError: input must be in the image of the Drinfeld module
```

```
>>> from sage.all import *
>>> t = phi.ore_polring().gen()
>>> phi.invert(t + Integer(1))
Traceback (most recent call last):
...
ValueError: input must be in the image of the Drinfeld module
```

```
sage: phi.invert(t^4 + t^2 + 1)
Traceback (most recent call last):
...
ValueError: input must be in the image of the Drinfeld module
```

```
>>> from sage.all import *
>>> phi.invert(t**Integer(4) + t**Integer(2) + Integer(1))
Traceback (most recent call last):
...
ValueError: input must be in the image of the Drinfeld module
```

ALGORITHM:

The algorithm relies on the inversion of a linear algebra system. See [MS2019], 3.2.5 for details.

is_isogenous (*psi*)

Return True when self is isogenous to the other Drinfeld module.

If the Drinfeld modules do not belong to the same category, an exception is raised.

EXAMPLES:

```
sage: Fq = GF(2)
sage: A.<T> = Fq[]
sage: K.<z> = Fq.extension(3)
sage: psi = DrinfeldModule(A, [z, z + 1, z^2 + z + 1])
sage: phi = DrinfeldModule(A, [z, z^2 + z + 1, z^2 + z])
sage: phi.is_isogenous(psi)
True
```

```
>>> from sage.all import *
>>> Fq = GF(Integer(2))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(3), names=('z',)); (z,) = K._first_ngens(1)
>>> psi = DrinfeldModule(A, [z, z + Integer(1), z**Integer(2) + z +
↪ Integer(1)])
>>> phi = DrinfeldModule(A, [z, z**Integer(2) + z + Integer(1), z**Integer(2)
↪ + z])
>>> phi.is_isogenous(psi)
True
```

```
sage: chi = DrinfeldModule(A, [z, z + 1, z^2 + z])
sage: phi.is_isogenous(chi)
False
```

```
>>> from sage.all import *
>>> chi = DrinfeldModule(A, [z, z + Integer(1), z**Integer(2) + z])
>>> phi.is_isogenous(chi)
False
```

```
sage: mu = DrinfeldModule(A, [z + 1, z^2 + z + 1, z^2 + z])
sage: phi.is_isogenous(mu)
Traceback (most recent call last):
...
TypeError: Drinfeld modules are not in the same category
```

```
>>> from sage.all import *
>>> mu = DrinfeldModule(A, [z + Integer(1), z**Integer(2) + z + Integer(1),
↳ z**Integer(2) + z])
>>> phi.is_isogenous(mu)
Traceback (most recent call last):
...
TypeError: Drinfeld modules are not in the same category
```

```
sage: mu = 1
sage: phi.is_isogenous(mu)
Traceback (most recent call last):
...
TypeError: input must be a Drinfeld module
```

```
>>> from sage.all import *
>>> mu = Integer(1)
>>> phi.is_isogenous(mu)
Traceback (most recent call last):
...
TypeError: input must be a Drinfeld module
```

ALGORITHM:

Two Drinfeld A-modules of equal characteristic are isogenous if and only if:

- they have the same rank
- the characteristic polynomial of the Frobenius endomorphism for both Drinfeld modules are equal.

is_ordinary()

Return True if this Drinfeld module is ordinary.

A Drinfeld module is ordinary if and only if its height is one.

EXAMPLES:

```
sage: Fq = GF(343)
sage: A.<T> = Fq[]
sage: K.<z6> = Fq.extension(2)
sage: phi = DrinfeldModule(A, [1, 0, z6])
sage: phi.is_ordinary()
False
```

```
>>> from sage.all import *
>>> Fq = GF(Integer(343))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(2), names=('z6',)); (z6,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [Integer(1), Integer(0), z6])
>>> phi.is_ordinary()
False
```

```
sage: phi = DrinfeldModule(A, [1, z6, 0, z6])
sage: phi.is_ordinary()
True
```

```
>>> from sage.all import *
>>> phi = DrinfeldModule(A, [Integer(1), z6, Integer(0), z6])
>>> phi.is_ordinary()
True
```

is_supersingular()

Return True if this Drinfeld module is supersingular.

A Drinfeld module is supersingular if and only if its height equals its rank.

EXAMPLES:

```
sage: Fq = GF(343)
sage: A.<T> = Fq[]
sage: K.<z6> = Fq.extension(2)
sage: phi = DrinfeldModule(A, [1, 0, z6])
sage: phi.is_supersingular()
True
sage: phi(phi.characteristic())    # Purely inseparable
z6*t^2
```

```
>>> from sage.all import *
>>> Fq = GF(Integer(343))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(2), names=('z6',)); (z6,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [Integer(1), Integer(0), z6])
>>> phi.is_supersingular()
True
>>> phi(phi.characteristic())    # Purely inseparable
z6*t^2
```

In rank two, a Drinfeld module is either ordinary or supersingular. In higher ranks, it could be neither of the two:

```
sage: psi = DrinfeldModule(A, [1, 0, z6, z6])
sage: psi.is_ordinary()
False
sage: psi.is_supersingular()
False
```

```
>>> from sage.all import *
>>> psi = DrinfeldModule(A, [Integer(1), Integer(0), z6, z6])
>>> psi.is_ordinary()
False
>>> psi.is_supersingular()
False
```


MORPHISMS AND ISOGENIES

2.1 Drinfeld module morphisms

This module provides the class `sage.rings.function_fields.drinfeld_module.morphism.DrinfeldModuleMorphism`.

AUTHORS: - Antoine Leudière (2022-04)

class `sage.rings.function_field.drinfeld_modules.morphism.DrinfeldModuleMorphism` (*parent, ore_pol*)

Bases: `Morphism`, `UniqueRepresentation`

This class represents Drinfeld $\mathbb{F}_q[T]$ -module morphisms.

Let ϕ and ψ be two Drinfeld $\mathbb{F}_q[T]$ -modules over a field K . A *morphism of Drinfeld modules* $\phi \rightarrow \psi$ is an Ore polynomial $f \in K\{\tau\}$ such that $f\phi_a = \psi_a f$ for every $a \in \mathbb{F}_q[T]$. In our case, this is equivalent to $f\phi_T = \psi_T f$. An *isogeny* is a nonzero morphism.

To create a morphism object, the user should never explicitly instantiate `DrinfeldModuleMorphism`, but rather call the parent homset with the defining Ore polynomial:

```
sage: Fq = GF(4)
sage: A.<T> = Fq[]
sage: K.<z> = Fq.extension(3)
sage: phi = DrinfeldModule(A, [z, z^2 + z, z^2 + z])
sage: t = phi.ore_polring().gen()
sage: ore_pol = t + z^5 + z^3 + z + 1
sage: psi = phi.velu(ore_pol)
sage: morphism = Hom(phi, psi)(ore_pol)
sage: morphism
Drinfeld Module morphism:
  From: Drinfeld module defined by T |--> (z^2 + z)*t^2 + (z^2 + z)*t + z
  To:   Drinfeld module defined by T |--> (z^5 + z^2 + z + 1)*t^2 + (z^4 + z + 1)*t + z
  Defn: t + z^5 + z^3 + z + 1
```

```
>>> from sage.all import *
>>> Fq = GF(Integer(4))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(3), names=('z',)); (z,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [z, z**Integer(2) + z, z**Integer(2) + z])
>>> t = phi.ore_polring().gen()
>>> ore_pol = t + z**Integer(5) + z**Integer(3) + z + Integer(1)
```

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```
>>> psi = phi.velu(ore_pol)
>>> morphism = Hom(phi, psi)(ore_pol)
>>> morphism
Drinfeld Module morphism:
  From: Drinfeld module defined by T |--> (z^2 + z)*t^2 + (z^2 + z)*t + z
  To:   Drinfeld module defined by T |--> (z^5 + z^2 + z + 1)*t^2 + (z^4 + z + 1)*t + z
  Defn: t + z^5 + z^3 + z + 1
```

The given Ore polynomial must indeed define a morphism:

```
sage: morphism = Hom(phi, psi)(1)
Traceback (most recent call last):
...
ValueError: Ore polynomial does not define a morphism
```

```
>>> from sage.all import *
>>> morphism = Hom(phi, psi)(Integer(1))
Traceback (most recent call last):
...
ValueError: Ore polynomial does not define a morphism
```

One can get basic data on the morphism:

```
sage: morphism.domain()
Drinfeld module defined by T |--> (z^2 + z)*t^2 + (z^2 + z)*t + z
sage: morphism.domain() is phi
True

sage: morphism.codomain()
Drinfeld module defined by T |--> (z^5 + z^2 + z + 1)*t^2 + (z^4 + z + 1)*t + z
sage: morphism.codomain() is psi
True
```

```
>>> from sage.all import *
>>> morphism.domain()
Drinfeld module defined by T |--> (z^2 + z)*t^2 + (z^2 + z)*t + z
>>> morphism.domain() is phi
True

>>> morphism.codomain()
Drinfeld module defined by T |--> (z^5 + z^2 + z + 1)*t^2 + (z^4 + z + 1)*t + z
>>> morphism.codomain() is psi
True
```

```
sage: morphism.ore_polynomial()
t + z^5 + z^3 + z + 1
sage: morphism.ore_polynomial() is ore_pol
True
```

```
>>> from sage.all import *
>>> morphism.ore_polynomial()
t + z^5 + z^3 + z + 1
>>> morphism.ore_polynomial() is ore_pol
True
```

One can check various properties:

```
sage: morphism.is_zero()
False
sage: morphism.is_isogeny()
True
sage: morphism.is_endomorphism()
False
sage: morphism.is_isomorphism()
False
```

```
>>> from sage.all import *
>>> morphism.is_zero()
False
>>> morphism.is_isogeny()
True
>>> morphism.is_endomorphism()
False
>>> morphism.is_isomorphism()
False
```

characteristic_polynomial (*var*='X')

Return the characteristic polynomial of this endomorphism.

INPUT:

- *var* – a string (default: X), the name of the variable of the characteristic polynomial

EXAMPLES:

```
sage: Fq = GF(5)
sage: A.<T> = Fq[]
sage: K.<z> = Fq.extension(3)
sage: phi = DrinfeldModule(A, [z, 0, 1, z])

sage: f = phi.frobenius_endomorphism()
sage: f.characteristic_polynomial()
X^3 + (T + 1)*X^2 + (2*T + 3)*X + 2*T^3 + T + 1
```

```
>>> from sage.all import *
>>> Fq = GF(Integer(5))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(3), names=('z',)); (z,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [z, Integer(0), Integer(1), z])

>>> f = phi.frobenius_endomorphism()
>>> f.characteristic_polynomial()
X^3 + (T + 1)*X^2 + (2*T + 3)*X + 2*T^3 + T + 1
```

We verify, on an example, that the characteristic polynomial of a morphism corresponding to ϕ_a is $(X - a)^r$ where r is the rank:

```
sage: g = phi.hom(T^2 + 1)
sage: chi = g.characteristic_polynomial()
sage: chi.factor()
(X + 4*T^2 + 4)^3
```

```
>>> from sage.all import *
>>> g = phi.hom(T**Integer(2) + Integer(1))
>>> chi = g.characteristic_polynomial()
>>> chi.factor()
(X + 4*T^2 + 4)^3
```

An example with another variable name:

```
sage: f.characteristic_polynomial(var='Y')
Y^3 + (T + 1)*Y^2 + (2*T + 3)*Y + 2*T^3 + T + 1
```

```
>>> from sage.all import *
>>> f.characteristic_polynomial(var='Y')
Y^3 + (T + 1)*Y^2 + (2*T + 3)*Y + 2*T^3 + T + 1
```

charpoly(var='X')

Return the characteristic polynomial of this endomorphism.

INPUT:

- var – a string (default: X), the name of the variable of the characteristic polynomial

EXAMPLES:

```
sage: Fq = GF(5)
sage: A.<T> = Fq[]
sage: K.<z> = Fq.extension(3)
sage: phi = DrinfeldModule(A, [z, 0, 1, z])

sage: f = phi.frobenius_endomorphism()
sage: chi = f.charpoly()
sage: chi
X^3 + (T + 1)*X^2 + (2*T + 3)*X + 2*T^3 + T + 1
```

```
>>> from sage.all import *
>>> Fq = GF(Integer(5))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(3), names=('z',)); (z,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [z, Integer(0), Integer(1), z])

>>> f = phi.frobenius_endomorphism()
>>> chi = f.charpoly()
>>> chi
X^3 + (T + 1)*X^2 + (2*T + 3)*X + 2*T^3 + T + 1
```

We check that the characteristic polynomial annihilates the morphism (Cayley-Hamilton's theorem):

```
sage: chi(f)
Endomorphism of Drinfeld module defined by T |--> z*t^3 + t^2 + z
Defn: 0
```

```
>>> from sage.all import *
>>> chi(f)
Endomorphism of Drinfeld module defined by T |--> z*t^3 + t^2 + z
Defn: 0
```

We verify, on an example, that the characteristic polynomial of the morphism corresponding to ϕ_a is $(X - a)^r$ where r is the rank:

```
sage: g = phi.hom(T^2 + 1)
sage: g.charpoly().factor()
(X + 4*T^2 + 4)^3
```

```
>>> from sage.all import *
>>> g = phi.hom(T**Integer(2) + Integer(1))
>>> g.charpoly().factor()
(X + 4*T^2 + 4)^3
```

An example with another variable name:

```
sage: f.charpoly(var='Y')
Y^3 + (T + 1)*Y^2 + (2*T + 3)*Y + 2*T^3 + T + 1
```

```
>>> from sage.all import *
>>> f.charpoly(var='Y')
Y^3 + (T + 1)*Y^2 + (2*T + 3)*Y + 2*T^3 + T + 1
```

`dual_isogeny()`

Return a dual isogeny to this morphism.

By definition, a dual isogeny of $f : \phi \rightarrow \psi$ is an isogeny $g : \psi \rightarrow \phi$ such that the composite $g \circ f$ is the multiplication by a generator of the norm of f .

EXAMPLES:

```
sage: Fq = GF(5)
sage: A.<T> = Fq[]
sage: K.<z> = Fq.extension(3)
sage: phi = DrinfeldModule(A, [z, 0, 1, z])
sage: t = phi.ore_variable()
sage: f = phi.hom(t + 1)
sage: f
Drinfeld Module morphism:
  From: Drinfeld module defined by T |--> z*t^3 + t^2 + z
  To:   Drinfeld module defined by T |--> (2*z^2 + 4*z + 4)*t^3 + (3*z^2 +
↪ 2*z + 2)*t^2 + (2*z^2 + 3*z + 4)*t + z
  Defn: t + 1
sage: g = f.dual_isogeny()
sage: g
Drinfeld Module morphism:
  From: Drinfeld module defined by T |--> (2*z^2 + 4*z + 4)*t^3 + (3*z^2 +
↪ 2*z + 2)*t^2 + (2*z^2 + 3*z + 4)*t + z
  To:   Drinfeld module defined by T |--> z*t^3 + t^2 + z
  Defn: z*t^2 + (4*z + 1)*t + z + 4
```

```
>>> from sage.all import *
>>> Fq = GF(Integer(5))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(3), names=('z',)); (z,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [z, Integer(0), Integer(1), z])
>>> t = phi.ore_variable()
>>> f = phi.hom(t + Integer(1))
>>> f
Drinfeld Module morphism:
  From: Drinfeld module defined by T |--> z*t^3 + t^2 + z
```

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```

To:   Drinfeld module defined by T |--> (2*z^2 + 4*z + 4)*t^3 + (3*z^2 +
↪2*z + 2)*t^2 + (2*z^2 + 3*z + 4)*t + z
Defn: t + 1
>>> g = f.dual_isogeny()
>>> g
Drinfeld Module morphism:
From: Drinfeld module defined by T |--> (2*z^2 + 4*z + 4)*t^3 + (3*z^2 +
↪2*z + 2)*t^2 + (2*z^2 + 3*z + 4)*t + z
To:   Drinfeld module defined by T |--> z*t^3 + t^2 + z
Defn: z*t^2 + (4*z + 1)*t + z + 4

```

We check that $f \circ g$ (resp. $g \circ f$) is the multiplication by the norm of f :

```

sage: a = f.norm().gen(); a
T + 4
sage: g * f == phi.hom(a)
True

```

```

sage: psi = f.codomain()
sage: f * g == psi.hom(a)
True

```

```

>>> from sage.all import *
>>> a = f.norm().gen(); a
T + 4
>>> g * f == phi.hom(a)
True

>>> psi = f.codomain()
>>> f * g == psi.hom(a)
True

```

inverse()

Return the inverse of this morphism.

Only morphisms defined by constant nonzero Ore polynomials are invertible.

EXAMPLES:

```

sage: Fq = GF(5)
sage: A.<T> = Fq[]
sage: K.<z> = Fq.extension(3)
sage: phi = DrinfeldModule(A, [z, 1, z, z^2])
sage: f = phi.hom(2); f
Endomorphism of Drinfeld module defined by T |--> z^2*t^3 + z*t^2 + t + z
Defn: 2
sage: f.inverse()
Endomorphism of Drinfeld module defined by T |--> z^2*t^3 + z*t^2 + t + z
Defn: 3

```

```

>>> from sage.all import *
>>> Fq = GF(Integer(5))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(3), names=('z',)); (z,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [z, Integer(1), z, z**Integer(2)])
>>> f = phi.hom(Integer(2)); f

```

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```

Endomorphism of Drinfeld module defined by T |--> z^2*t^3 + z*t^2 + t + z
Defn: 2
>>> f.inverse()
Endomorphism of Drinfeld module defined by T |--> z^2*t^3 + z*t^2 + t + z
Defn: 3

```

Inversion of general isomorphisms between different Drinfeld modules also works:

```

sage: g = phi.hom(z); g
Drinfeld Module morphism:
  From: Drinfeld module defined by T |--> z^2*t^3 + z*t^2 + t + z
  To:   Drinfeld module defined by T |--> z^2*t^3 + (z^2 + 2*z + 3)*t^2 + (z^
↪2 + 3*z)*t + z
  Defn: z
sage: g.inverse()
Drinfeld Module morphism:
  From: Drinfeld module defined by T |--> z^2*t^3 + (z^2 + 2*z + 3)*t^2 + (z^
↪2 + 3*z)*t + z
  To:   Drinfeld module defined by T |--> z^2*t^3 + z*t^2 + t + z
  Defn: 3*z^2 + 4

```

```

>>> from sage.all import *
>>> g = phi.hom(z); g
Drinfeld Module morphism:
  From: Drinfeld module defined by T |--> z^2*t^3 + z*t^2 + t + z
  To:   Drinfeld module defined by T |--> z^2*t^3 + (z^2 + 2*z + 3)*t^2 + (z^
↪2 + 3*z)*t + z
  Defn: z
>>> g.inverse()
Drinfeld Module morphism:
  From: Drinfeld module defined by T |--> z^2*t^3 + (z^2 + 2*z + 3)*t^2 + (z^
↪2 + 3*z)*t + z
  To:   Drinfeld module defined by T |--> z^2*t^3 + z*t^2 + t + z
  Defn: 3*z^2 + 4

```

When the morphism is not invertible, an error is raised:

```

sage: F = phi.frobenius_endomorphism()
sage: F.inverse()
Traceback (most recent call last):
...
ZeroDivisionError: this morphism is not invertible

```

```

>>> from sage.all import *
>>> F = phi.frobenius_endomorphism()
>>> F.inverse()
Traceback (most recent call last):
...
ZeroDivisionError: this morphism is not invertible

```

is_identity()

Return True whether the morphism is the identity morphism.

EXAMPLES:

```

sage: Fq = GF(2)
sage: A.<T> = Fq[]
sage: K.<z6> = Fq.extension(6)
sage: phi = DrinfeldModule(A, [z6, 1, 1])
sage: morphism = End(phi)(1)
sage: morphism.is_identity()
True

```

```

>>> from sage.all import *
>>> Fq = GF(Integer(2))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(6), names=('z6',)); (z6,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [z6, Integer(1), Integer(1)])
>>> morphism = End(phi)(Integer(1))
>>> morphism.is_identity()
True

```

```

sage: psi = DrinfeldModule(A, [z6, z6^4 + z6^2 + 1, 1])
sage: t = phi.ore_polring().gen()
sage: morphism = Hom(phi, psi)(t + z6^5 + z6^2 + 1)
sage: morphism.is_identity()
False

```

```

>>> from sage.all import *
>>> psi = DrinfeldModule(A, [z6, z6**Integer(4) + z6**Integer(2) + Integer(1),
↪ Integer(1)])
>>> t = phi.ore_polring().gen()
>>> morphism = Hom(phi, psi)(t + z6**Integer(5) + z6**Integer(2) + Integer(1))
>>> morphism.is_identity()
False

```

`is_isogeny()`

Return True whether the morphism is an isogeny.

EXAMPLES:

```

sage: Fq = GF(2)
sage: A.<T> = Fq[]
sage: K.<z6> = Fq.extension(6)
sage: phi = DrinfeldModule(A, [z6, 1, 1])
sage: psi = DrinfeldModule(A, [z6, z6^4 + z6^2 + 1, 1])
sage: t = phi.ore_polring().gen()
sage: morphism = Hom(phi, psi)(t + z6^5 + z6^2 + 1)
sage: morphism.is_isogeny()
True

```

```

>>> from sage.all import *
>>> Fq = GF(Integer(2))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(6), names=('z6',)); (z6,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [z6, Integer(1), Integer(1)])
>>> psi = DrinfeldModule(A, [z6, z6**Integer(4) + z6**Integer(2) + Integer(1),
↪ Integer(1)])
>>> t = phi.ore_polring().gen()
>>> morphism = Hom(phi, psi)(t + z6**Integer(5) + z6**Integer(2) + Integer(1))

```

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```
>>> morphism.is_isogeny()
True
```

```
sage: zero_morphism = End(phi)(0)
sage: zero_morphism.is_isogeny()
False
```

```
>>> from sage.all import *
>>> zero_morphism = End(phi)(Integer(0))
>>> zero_morphism.is_isogeny()
False
```

```
sage: identity_morphism = End(phi)(1)
sage: identity_morphism.is_isogeny()
True
```

```
>>> from sage.all import *
>>> identity_morphism = End(phi)(Integer(1))
>>> identity_morphism.is_isogeny()
True
```

```
sage: frobenius_endomorphism = phi.frobenius_endomorphism()
sage: frobenius_endomorphism.is_isogeny()
True
```

```
>>> from sage.all import *
>>> frobenius_endomorphism = phi.frobenius_endomorphism()
>>> frobenius_endomorphism.is_isogeny()
True
```

is_isomorphism()

Return True whether the morphism is an isomorphism.

EXAMPLES:

```
sage: Fq = GF(2)
sage: A.<T> = Fq[]
sage: K.<z6> = Fq.extension(6)
sage: phi = DrinfeldModule(A, [z6, 1, 1])
sage: psi = DrinfeldModule(A, [z6, z6^4 + z6^2 + 1, 1])
sage: t = phi.ore_polring().gen()
sage: morphism = Hom(phi, psi)(t + z6^5 + z6^2 + 1)
sage: morphism.is_isomorphism()
False
```

```
>>> from sage.all import *
>>> Fq = GF(Integer(2))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(6), names=('z6',)); (z6,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [z6, Integer(1), Integer(1)])
>>> psi = DrinfeldModule(A, [z6, z6**Integer(4) + z6**Integer(2) + Integer(1),
↪ Integer(1)])
>>> t = phi.ore_polring().gen()
>>> morphism = Hom(phi, psi)(t + z6**Integer(5) + z6**Integer(2) + Integer(1))
```

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```
>>> morphism.is_isomorphism()
False
```

```
sage: zero_morphism = End(phi)(0)
sage: zero_morphism.is_isomorphism()
False
```

```
>>> from sage.all import *
>>> zero_morphism = End(phi)(Integer(0))
>>> zero_morphism.is_isomorphism()
False
```

```
sage: identity_morphism = End(phi)(1)
sage: identity_morphism.is_isomorphism()
True
```

```
>>> from sage.all import *
>>> identity_morphism = End(phi)(Integer(1))
>>> identity_morphism.is_isomorphism()
True
```

```
sage: frobenius_endomorphism = phi.frobenius_endomorphism()
sage: frobenius_endomorphism.is_isomorphism()
False
```

```
>>> from sage.all import *
>>> frobenius_endomorphism = phi.frobenius_endomorphism()
>>> frobenius_endomorphism.is_isomorphism()
False
```

is_zero()

Return True whether the morphism is the zero morphism.

EXAMPLES:

```
sage: Fq = GF(2)
sage: A.<T> = Fq[]
sage: K.<z6> = Fq.extension(6)
sage: phi = DrinfeldModule(A, [z6, 1, 1])
sage: psi = DrinfeldModule(A, [z6, z6^4 + z6^2 + 1, 1])
sage: t = phi.ore_polring().gen()
sage: morphism = Hom(phi, psi)(t + z6^5 + z6^2 + 1)
sage: morphism.is_zero()
False
```

```
>>> from sage.all import *
>>> Fq = GF(Integer(2))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(6), names=('z6',)); (z6,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [z6, Integer(1), Integer(1)])
>>> psi = DrinfeldModule(A, [z6, z6**Integer(4) + z6**Integer(2) + Integer(1),
↪ Integer(1)])
>>> t = phi.ore_polring().gen()
>>> morphism = Hom(phi, psi)(t + z6**Integer(5) + z6**Integer(2) + Integer(1))
```

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```
>>> morphism.is_zero()
False
```

```
sage: zero_morphism = End(phi)(0)
sage: zero_morphism.is_zero()
True
```

```
>>> from sage.all import *
>>> zero_morphism = End(phi)(Integer(0))
>>> zero_morphism.is_zero()
True
```

norm (*ideal=True*)

Return the norm of this isogeny.

INPUT:

- *ideal* – a boolean (default: True); if True, return the norm as an ideal in the function ring of the Drinfeld modules; if False, return the norm as an element in this function ring (only relevant for endomorphisms)

EXAMPLES:

```
sage: Fq = GF(5)
sage: A.<T> = Fq[]
sage: K.<z> = Fq.extension(3)
sage: phi = DrinfeldModule(A, [z, 0, 1, z])
sage: t = phi.ore_variable()
sage: f = phi.hom(t + 1)
sage: f.norm()
Principal ideal (T + 4) of Univariate Polynomial Ring in T over Finite Field_
↳ of size 5
```

```
>>> from sage.all import *
>>> Fq = GF(Integer(5))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(3), names=('z',)); (z,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [z, Integer(0), Integer(1), z])
>>> t = phi.ore_variable()
>>> f = phi.hom(t + Integer(1))
>>> f.norm()
Principal ideal (T + 4) of Univariate Polynomial Ring in T over Finite Field_
↳ of size 5
```

The norm of the Frobenius endomorphism is equal to the characteristic:

```
sage: F = phi.frobenius_endomorphism()
sage: F.norm()
Principal ideal (T^3 + 3*T + 3) of Univariate Polynomial Ring in T over_
↳ Finite Field of size 5
sage: phi.characteristic()
T^3 + 3*T + 3
```

```
>>> from sage.all import *
>>> F = phi.frobenius_endomorphism()
>>> F.norm()
```

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```
Principal ideal (T^3 + 3*T + 3) of Univariate Polynomial Ring in T over
↪ Finite Field of size 5
>>> phi.characteristic()
T^3 + 3*T + 3
```

For a in the underlying function ring, the norm of the endomorphism given by ϕ_a is a^r where r is the rank:

```
sage: g = phi.hom(T)
sage: g.norm()
Principal ideal (T^3) of Univariate Polynomial Ring in T over Finite Field of
↪ size 5

sage: h = phi.hom(T+1)
sage: h.norm()
Principal ideal (T^3 + 3*T^2 + 3*T + 1) of Univariate Polynomial Ring in T
↪ over Finite Field of size 5
```

```
>>> from sage.all import *
>>> g = phi.hom(T)
>>> g.norm()
Principal ideal (T^3) of Univariate Polynomial Ring in T over Finite Field of
↪ size 5

>>> h = phi.hom(T+Integer(1))
>>> h.norm()
Principal ideal (T^3 + 3*T^2 + 3*T + 1) of Univariate Polynomial Ring in T
↪ over Finite Field of size 5
```

For endomorphisms, the norm is not an ideal of A but it makes sense as an actual element of A . We can get this element by passing in the argument `ideal=False`:

```
sage: phi.hom(2*T).norm(ideal=False)
3*T^3

sage: f.norm(ideal=False)
Traceback (most recent call last):
...
ValueError: norm is defined as an actual element only for endomorphisms
```

```
>>> from sage.all import *
>>> phi.hom(Integer(2)*T).norm(ideal=False)
3*T^3

>>> f.norm(ideal=False)
Traceback (most recent call last):
...
ValueError: norm is defined as an actual element only for endomorphisms
```

`ore_polynomial()`

Return the Ore polynomial that defines the morphism.

EXAMPLES:

```
sage: Fq = GF(2)
sage: A.<T> = Fq[]
sage: K.<z6> = Fq.extension(6)
```

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```

sage: phi = DrinfeldModule(A, [z6, 1, 1])
sage: psi = DrinfeldModule(A, [z6, z6^4 + z6^2 + 1, 1])
sage: t = phi.ore_polring().gen()
sage: morphism = Hom(phi, psi)(t + z6^5 + z6^2 + 1)
sage: ore_pol = morphism.ore_polynomial()
sage: ore_pol
t + z6^5 + z6^2 + 1

```

```

>>> from sage.all import *
>>> Fq = GF(Integer(2))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(6), names=('z6',)); (z6,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [z6, Integer(1), Integer(1)])
>>> psi = DrinfeldModule(A, [z6, z6**Integer(4) + z6**Integer(2) + Integer(1),
↪ Integer(1)])
>>> t = phi.ore_polring().gen()
>>> morphism = Hom(phi, psi)(t + z6**Integer(5) + z6**Integer(2) + Integer(1))
>>> ore_pol = morphism.ore_polynomial()
>>> ore_pol
t + z6^5 + z6^2 + 1

```

```

sage: ore_pol * phi(T) == psi(T) * ore_pol
True

```

```

>>> from sage.all import *
>>> ore_pol * phi(T) == psi(T) * ore_pol
True

```

2.2 Set of morphisms between two Drinfeld modules

This module provides the class `sage.rings.function_field.drinfeld_module.homset.DrinfeldModuleHomset`.

AUTHORS:

- Antoine Leudière (2022-04)

```

class sage.rings.function_field.drinfeld_modules.homset.DrinfeldModuleHomset(X,
                                                                              Y,
                                                                              cat-
                                                                              e-
                                                                              gory=None,
                                                                              check=True)

```

Bases: `Homset`

This class implements the set of morphisms between two Drinfeld $\mathbb{F}_q[T]$ -modules.

INPUT:

- X – the domain
- Y – the codomain

EXAMPLES:

```

sage: Fq = GF(27)
sage: A.<T> = Fq[]
sage: K.<z6> = Fq.extension(2)
sage: phi = DrinfeldModule(A, [z6, z6, 2])
sage: psi = DrinfeldModule(A, [z6, 2*z6^5 + 2*z6^4 + 2*z6 + 1, 2])
sage: H = Hom(phi, psi)
sage: H
Set of Drinfeld module morphisms
  from (gen) 2*t^2 + z6*t + z6
  to (gen) 2*t^2 + (2*z6^5 + 2*z6^4 + 2*z6 + 1)*t + z6

```

```

>>> from sage.all import *
>>> Fq = GF(Integer(27))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(2), names=('z6',)); (z6,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [z6, z6, Integer(2)])
>>> psi = DrinfeldModule(A, [z6, Integer(2)*z6**Integer(5) +
↳ Integer(2)*z6**Integer(4) + Integer(2)*z6 + Integer(1), Integer(2)])
>>> H = Hom(phi, psi)
>>> H
Set of Drinfeld module morphisms
  from (gen) 2*t^2 + z6*t + z6
  to (gen) 2*t^2 + (2*z6^5 + 2*z6^4 + 2*z6 + 1)*t + z6

```

```

sage: from sage.rings.function_field.drinfeld_modules.homset import
↳ DrinfeldModuleHomset
sage: isinstance(H, DrinfeldModuleHomset)
True

```

```

>>> from sage.all import *
>>> from sage.rings.function_field.drinfeld_modules.homset import
↳ DrinfeldModuleHomset
>>> isinstance(H, DrinfeldModuleHomset)
True

```

There is a simpler syntax for endomorphisms sets:

```

sage: E = End(phi)
sage: E
Set of Drinfeld module morphisms from (gen) 2*t^2 + z6*t + z6 to (gen) 2*t^2 +
↳ z6*t + z6
sage: E is Hom(phi, phi)
True

```

```

>>> from sage.all import *
>>> E = End(phi)
>>> E
Set of Drinfeld module morphisms from (gen) 2*t^2 + z6*t + z6 to (gen) 2*t^2 +
↳ z6*t + z6
>>> E is Hom(phi, phi)
True

```

The domain and codomain must have the same Drinfeld modules category:

```

sage: rho = DrinfeldModule(A, [Frac(A)(T), 1])
sage: Hom(phi, rho)

```

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```
Traceback (most recent call last):
...
ValueError: Drinfeld modules must be in the same category
```

```
>>> from sage.all import *
>>> rho = DrinfeldModule(A, [Frac(A)(T), Integer(1)])
>>> Hom(phi, rho)
Traceback (most recent call last):
...
ValueError: Drinfeld modules must be in the same category
```

```
sage: sigma = DrinfeldModule(A, [1, z6, 2])
sage: Hom(phi, sigma)
Traceback (most recent call last):
...
ValueError: Drinfeld modules must be in the same category
```

```
>>> from sage.all import *
>>> sigma = DrinfeldModule(A, [Integer(1), z6, Integer(2)])
>>> Hom(phi, sigma)
Traceback (most recent call last):
...
ValueError: Drinfeld modules must be in the same category
```

One can create morphism objects by calling the homset:

```
sage: identity_morphism = E(1)
sage: identity_morphism
Identity morphism of Drinfeld module defined by  $T \mapsto 2t^2 + z6t + z6$ 
```

```
>>> from sage.all import *
>>> identity_morphism = E(Integer(1))
>>> identity_morphism
Identity morphism of Drinfeld module defined by  $T \mapsto 2t^2 + z6t + z6$ 
```

```
sage: t = phi.ore_polring().gen()
sage: frobenius_endomorphism = E(t^6)
sage: frobenius_endomorphism
Endomorphism of Drinfeld module defined by  $T \mapsto 2t^2 + z6t + z6$ 
Defn:  $t^6$ 
```

```
>>> from sage.all import *
>>> t = phi.ore_polring().gen()
>>> frobenius_endomorphism = E(t**Integer(6))
>>> frobenius_endomorphism
Endomorphism of Drinfeld module defined by  $T \mapsto 2t^2 + z6t + z6$ 
Defn:  $t^6$ 
```

```
sage: isogeny = H(t + 1)
sage: isogeny
Drinfeld Module morphism:
  From: Drinfeld module defined by  $T \mapsto 2t^2 + z6t + z6$ 
  To:   Drinfeld module defined by  $T \mapsto 2t^2 + (2z6^5 + 2z6^4 + 2z6 + 1)t$ 
```

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```
↪+ z6
Defn: t + 1
```

```
>>> from sage.all import *
>>> isogeny = H(t + Integer(1))
>>> isogeny
Drinfeld Module morphism:
  From: Drinfeld module defined by T |--> 2*t^2 + z6*t + z6
  To:   Drinfeld module defined by T |--> 2*t^2 + (2*z6^5 + 2*z6^4 + 2*z6 + 1)*t
↪+ z6
Defn: t + 1
```

And one can test if an Ore polynomial defines a morphism using the `in` syntax:

```
sage: 1 in H
False
sage: t^6 in H
False
sage: t + 1 in H
True
sage: 1 in E
True
sage: t^6 in E
True
sage: t + 1 in E
False
```

```
>>> from sage.all import *
>>> Integer(1) in H
False
>>> t**Integer(6) in H
False
>>> t + Integer(1) in H
True
>>> Integer(1) in E
True
>>> t**Integer(6) in E
True
>>> t + Integer(1) in E
False
```

This also works if the candidate is a morphism object:

```
sage: isogeny in H
True
sage: E(0) in E
True
sage: identity_morphism in H
False
sage: frobenius_endomorphism in H
False
```

```
>>> from sage.all import *
>>> isogeny in H
True
```

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```
>>> E(Integer(0)) in E
True
>>> identity_morphism in H
False
>>> frobenius_endomorphism in H
False
```

Elementalias of *DrinfeldModuleMorphism*

```
class sage.rings.function_field.drinfeld_modules.homset.DrinfeldModuleMorphismAction(A,
H,
is_left,
op)
```

Bases: *Action*

Action of the function ring on the homset of a Drinfeld module.

EXAMPLES:

```
sage: Fq = GF(5)
sage: A.<T> = Fq[]
sage: K.<z> = Fq.extension(3)
sage: phi = DrinfeldModule(A, [z, 1, z])
sage: psi = DrinfeldModule(A, [z, z^2 + 4*z + 3, 2*z^2 + 4*z + 4])
sage: H = Hom(phi, psi)
sage: t = phi.ore_variable()
sage: f = H(t + 2)
```

```
>>> from sage.all import *
>>> Fq = GF(Integer(5))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(3), names=('z',)); (z,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [z, Integer(1), z])
>>> psi = DrinfeldModule(A, [z, z**Integer(2) + Integer(4)*z + Integer(3),
↳ Integer(2)*z**Integer(2) + Integer(4)*z + Integer(4)])
>>> H = Hom(phi, psi)
>>> t = phi.ore_variable()
>>> f = H(t + Integer(2))
```

Left action:

```
sage: (T + 1) * f
Drinfeld Module morphism:
  From: Drinfeld module defined by T |--> z*t^2 + t + z
  To:   Drinfeld module defined by T |--> (2*z^2 + 4*z + 4)*t^2 + (z^2 + 4*z +
↳ 3)*t + z
  Defn: (2*z^2 + 4*z + 4)*t^3 + (2*z + 1)*t^2 + (2*z^2 + 4*z + 2)*t + 2*z + 2
```

```
>>> from sage.all import *
>>> (T + Integer(1)) * f
Drinfeld Module morphism:
  From: Drinfeld module defined by T |--> z*t^2 + t + z
  To:   Drinfeld module defined by T |--> (2*z^2 + 4*z + 4)*t^2 + (z^2 + 4*z +
↳ 3)*t + z
  Defn: (2*z^2 + 4*z + 4)*t^3 + (2*z + 1)*t^2 + (2*z^2 + 4*z + 2)*t + 2*z + 2
```

Right action currently does not work (it is a known bug, due to an incompatibility between multiplication of morphisms and the coercion system):

```
sage: f * (T + 1)
Traceback (most recent call last):
...
TypeError: right (=T + 1) must be a map to multiply it by Drinfeld Module
↪morphism:
  From: Drinfeld module defined by T |--> z*t^2 + t + z
  To:   Drinfeld module defined by T |--> (2*z^2 + 4*z + 4)*t^2 + (z^2 + 4*z +
↪3)*t + z
  Defn: t + 2
```

```
>>> from sage.all import *
>>> f * (T + Integer(1))
Traceback (most recent call last):
...
TypeError: right (=T + 1) must be a map to multiply it by Drinfeld Module
↪morphism:
  From: Drinfeld module defined by T |--> z*t^2 + t + z
  To:   Drinfeld module defined by T |--> (2*z^2 + 4*z + 4)*t^2 + (z^2 + 4*z +
↪3)*t + z
  Defn: t + 2
```

THE MODULE ACTION INDUCED BY A DRINFELD MODULE

3.1 The module action induced by a Drinfeld module

This module provides the class `sage.rings.function_field.drinfeld_module.action.DrinfeldModuleAction`.

AUTHORS:

- Antoine Leudière (2022-04)

class `sage.rings.function_field.drinfeld_modules.action.DrinfeldModuleAction` (*drinfeld_module*)

Bases: `Action`

This class implements the module action induced by a Drinfeld $\mathbb{F}_q[T]$ -module.

Let ϕ be a Drinfeld $\mathbb{F}_q[T]$ -module over a field K and let L/K be a field extension. Let $x \in L$ and let a be a function ring element; the action is defined as $(a, x) \mapsto \phi_a(x)$.

Note: In this implementation, L is K .

Note: The user should never explicitly instantiate the class *DrinfeldModuleAction*.

Warning: This class may be replaced later on. See issues #34833 and #34834.

INPUT: the Drinfeld module

EXAMPLES:

```
sage: Fq.<z2> = GF(11)
sage: A.<T> = Fq[]
sage: K.<z> = Fq.extension(2)
sage: phi = DrinfeldModule(A, [z, 0, 0, 1])
sage: action = phi.action()
sage: action
Action on Finite Field in z of size 11^2 over its base
induced by Drinfeld module defined by T |--> t^3 + z
```

```

>>> from sage.all import *
>>> Fq = GF(Integer(11), names=('z2',)); (z2,) = Fq._first_ngens(1)
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(2), names=('z',)); (z,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [z, Integer(0), Integer(0), Integer(1)])
>>> action = phi.action()
>>> action
Action on Finite Field in z of size 11^2 over its base
induced by Drinfeld module defined by T |--> t^3 + z

```

The action on elements is computed as follows:

```

sage: P = T + 1
sage: a = z
sage: action(P, a)
...
4*z + 2
sage: action(0, K.random_element())
0
sage: action(A.random_element(), 0)
0

```

```

>>> from sage.all import *
>>> P = T + Integer(1)
>>> a = z
>>> action(P, a)
...
4*z + 2
>>> action(Integer(0), K.random_element())
0
>>> action(A.random_element(), Integer(0))
0

```

Finally, given a Drinfeld module action, it is easy to recover the corresponding Drinfeld module:

```

sage: action.drinfeld_module() is phi
True

```

```

>>> from sage.all import *
>>> action.drinfeld_module() is phi
True

```

drinfeld_module()

Return the Drinfeld module defining the action.

OUTPUT: a Drinfeld module

EXAMPLES:

```

sage: Fq.<z2> = GF(11)
sage: A.<T> = Fq[]
sage: K.<z> = Fq.extension(2)
sage: phi = DrinfeldModule(A, [z, 0, 0, 1])
sage: action = phi.action()
sage: action.drinfeld_module() is phi
True

```

```
>>> from sage.all import *
>>> Fq = GF(Integer(11), names=('z2',)); (z2,) = Fq._first_ngens(1)
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(2), names=('z',)); (z,) = K._first_ngens(1)
>>> phi = DrinfeldModule(A, [z, Integer(0), Integer(0), Integer(1)])
>>> action = phi.action()
>>> action.drinfeld_module() is phi
True
```


THE CATEGORY OF DRINFELD MODULES

4.1 Drinfeld modules over a base

This module provides the class `sage.category.drinfeld_modules.DrinfeldModules`.

AUTHORS:

- Antoine Leudière (2022-04)
- Xavier Caruso (2022-06)

class `sage.categories.drinfeld_modules.DrinfeldModules` (*base_field*, *name='t'*)

Bases: `Category_over_base_ring`

This class implements the category of Drinfeld $\mathbb{F}_q[T]$ -modules on a given base field.

Let $\mathbb{F}_q[T]$ be a polynomial ring with coefficients in a finite field $\mathbb{F}_q$ and let K be a field. Fix a ring morphism $\gamma : \mathbb{F}_q[T] \rightarrow K$; we say that K is an $\mathbb{F}_q[T]$ -*field*. Let $K\{\tau\}$ be the ring of Ore polynomials with coefficients in K , whose multiplication is given by the rule $\tau\lambda = \lambda^q\tau$ for any $\lambda \in K$.

The extension $K/\mathbb{F}_q[T]$ (represented as an instance of the class `sage.rings.ring_extension.RingExtension`) is the *base field* of the category; its defining morphism γ is called the *base morphism*.

The monic polynomial that generates the kernel of γ is called the $\mathbb{F}_q[T]$ -*characteristic*, or *function-field characteristic*, of the base field. We say that $\mathbb{F}_q[T]$ is the *function ring* of the category; $K\{\tau\}$ is the *Ore polynomial ring*. The constant coefficient of the category is the image of T under the base morphism.

Construction

Generally, Drinfeld modules objects are created before their category, and the category is retrieved as an attribute of the Drinfeld module:

```
sage: Fq = GF(11)
sage: A.<T> = Fq[]
sage: K.<z> = Fq.extension(4)
sage: p_root = z^3 + 7*z^2 + 6*z + 10
sage: phi = DrinfeldModule(A, [p_root, 0, 0, 1])
sage: C = phi.category()
sage: C
Category of Drinfeld modules over Finite Field in z of size 11^4 over its base
```

```
>>> from sage.all import *
>>> Fq = GF(Integer(11))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
```

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```

>>> K = Fq.extension(Integer(4), names=('z',)); (z,) = K._first_ngens(1)
>>> p_root = z**Integer(3) + Integer(7)*z**Integer(2) + Integer(6)*z + Integer(10)
>>> phi = DrinfeldModule(A, [p_root, Integer(0), Integer(0), Integer(1)])
>>> C = phi.category()
>>> C
Category of Drinfeld modules over Finite Field in z of size 11^4 over its base

```

The output tells the user that the category is only defined by its base.

Properties of the category

The base field is retrieved using the method `base()`.

sage: C.base() Finite Field in z of size 11^4 over its base

Equivalently, one can use `base_morphism()` to retrieve the base morphism:

```

sage: C.base_morphism()
Ring morphism:
  From: Univariate Polynomial Ring in T over Finite Field of size 11
  To:   Finite Field in z of size 11^4 over its base
  Defn: T |--> z^3 + 7*z^2 + 6*z + 10

```

```

>>> from sage.all import *
>>> C.base_morphism()
Ring morphism:
  From: Univariate Polynomial Ring in T over Finite Field of size 11
  To:   Finite Field in z of size 11^4 over its base
  Defn: T |--> z^3 + 7*z^2 + 6*z + 10

```

The so-called constant coefficient — which is the same for all Drinfeld modules in the category — is simply the image of T by the base morphism:

```

sage: C.constant_coefficient()
z^3 + 7*z^2 + 6*z + 10
sage: C.base_morphism()(T) == C.constant_coefficient()
True

```

```

>>> from sage.all import *
>>> C.constant_coefficient()
z^3 + 7*z^2 + 6*z + 10
>>> C.base_morphism()(T) == C.constant_coefficient()
True

```

Similarly, the function ring-characteristic of the category is either 0 or the unique monic polynomial in $\mathbb{F}_q[T]$ that generates the kernel of the base:

```

sage: C.characteristic()
T^2 + 7*T + 2
sage: C.base_morphism()(C.characteristic())
0

```

```

>>> from sage.all import *
>>> C.characteristic()
T^2 + 7*T + 2

```

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```
>>> C.base_morphism()(C.characteristic())
0
```

The base field, base morphism, function ring and Ore polynomial ring are the same for the category and its objects:

```
sage: C.base() is phi.base()
True
sage: C.base_morphism() is phi.base_morphism()
True

sage: C.function_ring()
Univariate Polynomial Ring in T over Finite Field of size 11
sage: C.function_ring() is phi.function_ring()
True

sage: C.ore_polring()
Ore Polynomial Ring in t over Finite Field in z of size 11^4 over its base_
↪twisted by Frob
sage: C.ore_polring() is phi.ore_polring()
True
```

```
>>> from sage.all import *
>>> C.base() is phi.base()
True
>>> C.base_morphism() is phi.base_morphism()
True

>>> C.function_ring()
Univariate Polynomial Ring in T over Finite Field of size 11
>>> C.function_ring() is phi.function_ring()
True

>>> C.ore_polring()
Ore Polynomial Ring in t over Finite Field in z of size 11^4 over its base_
↪twisted by Frob
>>> C.ore_polring() is phi.ore_polring()
True
```

Creating Drinfeld module objects from the category

Calling `object()` with an Ore polynomial creates a Drinfeld module object in the category whose generator is the input:

```
sage: psi = C.object([p_root, 1])
sage: psi
Drinfeld module defined by T |--> t + z^3 + 7*z^2 + 6*z + 10
sage: psi.category() is C
True
```

```
>>> from sage.all import *
>>> psi = C.object([p_root, Integer(1)])
>>> psi
Drinfeld module defined by T |--> t + z^3 + 7*z^2 + 6*z + 10
>>> psi.category() is C
True
```

Of course, the constant coefficient of the input must be the same as the category:

```
sage: C.object([z, 1])
Traceback (most recent call last):
...
ValueError: constant coefficient must equal that of the category
```

```
>>> from sage.all import *
>>> C.object([z, Integer(1)])
Traceback (most recent call last):
...
ValueError: constant coefficient must equal that of the category
```

It is also possible to create a random object in the category. The input is the desired rank:

```
sage: rho = C.random_object(2)
sage: rho # random
Drinfeld module defined by  $T \mapsto (7z^3 + 7z^2 + 10z + 2)t^2 + (9z^3 + 5z^2 + 2z + 7)t + z^3 + 7z^2 + 6z + 10$ 
sage: rho.rank() == 2
True
sage: rho.category() is C
True
```

```
>>> from sage.all import *
>>> rho = C.random_object(Integer(2))
>>> rho # random
Drinfeld module defined by  $T \mapsto (7z^3 + 7z^2 + 10z + 2)t^2 + (9z^3 + 5z^2 + 2z + 7)t + z^3 + 7z^2 + 6z + 10$ 
>>> rho.rank() == Integer(2)
True
>>> rho.category() is C
True
```

Endsets()

Return the category of endsets.

EXAMPLES:

```
sage: Fq = GF(11)
sage: A.<T> = Fq[]
sage: K.<z> = Fq.extension(4)
sage: p_root = z^3 + 7*z^2 + 6*z + 10
sage: phi = DrinfeldModule(A, [p_root, 0, 0, 1])
sage: C = phi.category()

sage: from sage.categories.homsets import Homsets
sage: C.Endsets() is Homsets().Endsets()
True
```

```
>>> from sage.all import *
>>> Fq = GF(Integer(11))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(4), names=('z',)); (z,) = K._first_ngens(1)
>>> p_root = z**Integer(3) + Integer(7)*z**Integer(2) + Integer(6)*z + Integer(10)
>>> phi = DrinfeldModule(A, [p_root, Integer(0), Integer(0), Integer(1)])
```

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```
>>> C = phi.category()

>>> from sage.categories.homsets import Homsets
>>> C.Endsets() is Homsets().Endsets()
True
```

Homsets()

Return the category of homsets.

EXAMPLES:

```
sage: Fq = GF(11)
sage: A.<T> = Fq[]
sage: K.<z> = Fq.extension(4)
sage: p_root = z^3 + 7*z^2 + 6*z + 10
sage: phi = DrinfeldModule(A, [p_root, 0, 0, 1])
sage: C = phi.category()

sage: from sage.categories.homsets import Homsets
sage: C.Homsets() is Homsets()
True
```

```
>>> from sage.all import *
>>> Fq = GF(Integer(11))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(4), names=('z',)); (z,) = K._first_ngens(1)
>>> p_root = z**Integer(3) + Integer(7)*z**Integer(2) + Integer(6)*z + \
↳ Integer(10)
>>> phi = DrinfeldModule(A, [p_root, Integer(0), Integer(0), Integer(1)])
>>> C = phi.category()

>>> from sage.categories.homsets import Homsets
>>> C.Homsets() is Homsets()
True
```

class ParentMethods

Bases: object

base()

Return the base field of this Drinfeld module, viewed as an algebra over the function ring.

This is an instance of the class `sage.rings.ring_extension.RingExtension`.

EXAMPLES:

```
sage: Fq = GF(25)
sage: A.<T> = Fq[]
sage: K.<z12> = Fq.extension(6)
sage: p_root = 2*z12^11 + 2*z12^10 + z12^9 + 3*z12^8 + z12^7 + 2*z12^5 + \
↳ 2*z12^4 + 3*z12^3 + z12^2 + 2*z12
sage: phi = DrinfeldModule(A, [p_root, z12^3, z12^5])
sage: phi.base()
Finite Field in z12 of size 5^12 over its base
```

```
>>> from sage.all import *
>>> Fq = GF(Integer(25))
```

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```

>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(6), names=('z12',)); (z12,) = K._first_
↳ngens(1)
>>> p_root = Integer(2)*z12**Integer(11) + Integer(2)*z12**Integer(10) +
↳z12**Integer(9) + Integer(3)*z12**Integer(8) + z12**Integer(7) +
↳Integer(2)*z12**Integer(5) + Integer(2)*z12**Integer(4) +
↳Integer(3)*z12**Integer(3) + z12**Integer(2) + Integer(2)*z12
>>> phi = DrinfeldModule(A, [p_root, z12**Integer(3), z12**Integer(5)])
>>> phi.base()
Finite Field in z12 of size 5^12 over its base

```

The base can be infinite:

```

sage: sigma = DrinfeldModule(A, [Frac(A).gen(), 1])
sage: sigma.base()
Fraction Field of Univariate Polynomial Ring in T over Finite Field in z2
↳of size 5^2 over its base

```

```

>>> from sage.all import *
>>> sigma = DrinfeldModule(A, [Frac(A).gen(), Integer(1)])
>>> sigma.base()
Fraction Field of Univariate Polynomial Ring in T over Finite Field in z2
↳of size 5^2 over its base

```

base_morphism()

Return the base morphism of this Drinfeld module.

EXAMPLES:

```

sage: Fq = GF(25)
sage: A.<T> = Fq[]
sage: K.<z12> = Fq.extension(6)
sage: p_root = 2*z12^11 + 2*z12^10 + z12^9 + 3*z12^8 + z12^7 + 2*z12^5 +
↳2*z12^4 + 3*z12^3 + z12^2 + 2*z12
sage: phi = DrinfeldModule(A, [p_root, z12^3, z12^5])
sage: phi.base_morphism()
Ring morphism:
  From: Univariate Polynomial Ring in T over Finite Field in z2 of size 5^
↳2
  To:   Finite Field in z12 of size 5^12 over its base
  Defn: T |--> 2*z12^11 + 2*z12^10 + z12^9 + 3*z12^8 + z12^7 + 2*z12^5 +
↳2*z12^4 + 3*z12^3 + z12^2 + 2*z12

```

```

>>> from sage.all import *
>>> Fq = GF(Integer(25))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(6), names=('z12',)); (z12,) = K._first_
↳ngens(1)
>>> p_root = Integer(2)*z12**Integer(11) + Integer(2)*z12**Integer(10) +
↳z12**Integer(9) + Integer(3)*z12**Integer(8) + z12**Integer(7) +
↳Integer(2)*z12**Integer(5) + Integer(2)*z12**Integer(4) +
↳Integer(3)*z12**Integer(3) + z12**Integer(2) + Integer(2)*z12
>>> phi = DrinfeldModule(A, [p_root, z12**Integer(3), z12**Integer(5)])
>>> phi.base_morphism()
Ring morphism:
  From: Univariate Polynomial Ring in T over Finite Field in z2 of size 5^

```

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```

↪2
To: Finite Field in z12 of size 5^12 over its base
Defn: T |--> 2*z12^11 + 2*z12^10 + z12^9 + 3*z12^8 + z12^7 + 2*z12^5 +
↪2*z12^4 + 3*z12^3 + z12^2 + 2*z12

```

The base field can be infinite:

```

sage: sigma = DrinfeldModule(A, [Frac(A).gen(), 1])
sage: sigma.base_morphism()
Ring morphism:
  From: Univariate Polynomial Ring in T over Finite Field in z2 of size 5^
↪2
  To: Fraction Field of Univariate Polynomial Ring in T over Finite
↪Field in z2 of size 5^2 over its base
  Defn: T |--> T

```

```

>>> from sage.all import *
>>> sigma = DrinfeldModule(A, [Frac(A).gen(), Integer(1)])
>>> sigma.base_morphism()
Ring morphism:
  From: Univariate Polynomial Ring in T over Finite Field in z2 of size 5^
↪2
  To: Fraction Field of Univariate Polynomial Ring in T over Finite
↪Field in z2 of size 5^2 over its base
  Defn: T |--> T

```

base_over_constants_field()

Return the base field, seen as an extension over the constants field $\mathbb{F}_q$.

This is an instance of the class `sage.rings.ring_extension.RingExtension`.

EXAMPLES:

```

sage: Fq = GF(25)
sage: A.<T> = Fq[]
sage: K.<z12> = Fq.extension(6)
sage: p_root = 2*z12^11 + 2*z12^10 + z12^9 + 3*z12^8 + z12^7 + 2*z12^5 +
↪2*z12^4 + 3*z12^3 + z12^2 + 2*z12
sage: phi = DrinfeldModule(A, [p_root, z12^3, z12^5])
sage: phi.base_over_constants_field()
Field in z12 with defining polynomial x^6 + (4*z2 + 3)*x^5 + x^4 + (3*z2
↪+ 1)*x^3 + x^2 + (4*z2 + 1)*x + z2 over its base

```

```

>>> from sage.all import *
>>> Fq = GF(Integer(25))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(6), names=('z12',)); (z12,) = K._first_
↪ngens(1)
>>> p_root = Integer(2)*z12**Integer(11) + Integer(2)*z12**Integer(10) +
↪z12**Integer(9) + Integer(3)*z12**Integer(8) + z12**Integer(7) +
↪Integer(2)*z12**Integer(5) + Integer(2)*z12**Integer(4) +
↪Integer(3)*z12**Integer(3) + z12**Integer(2) + Integer(2)*z12
>>> phi = DrinfeldModule(A, [p_root, z12**Integer(3), z12**Integer(5)])
>>> phi.base_over_constants_field()
Field in z12 with defining polynomial x^6 + (4*z2 + 3)*x^5 + x^4 + (3*z2
↪+ 1)*x^3 + x^2 + (4*z2 + 1)*x + z2 over its base

```

characteristic()

Return the function ring-characteristic.

EXAMPLES:

```
sage: Fq = GF(25)
sage: A.<T> = Fq[]
sage: K.<z12> = Fq.extension(6)
sage: p_root = 2*z12^11 + 2*z12^10 + z12^9 + 3*z12^8 + z12^7 + 2*z12^5 +
↪ 2*z12^4 + 3*z12^3 + z12^2 + 2*z12
sage: phi = DrinfeldModule(A, [p_root, z12^3, z12^5])
sage: phi.characteristic()
T^2 + (4*z2 + 2)*T + 2
sage: phi.base_morphism()(phi.characteristic())
0
```

```
>>> from sage.all import *
>>> Fq = GF(Integer(25))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(6), names=('z12',)); (z12,) = K._first_
↪ ngens(1)
>>> p_root = Integer(2)*z12**Integer(11) + Integer(2)*z12**Integer(10) +
↪ z12**Integer(9) + Integer(3)*z12**Integer(8) + z12**Integer(7) +
↪ Integer(2)*z12**Integer(5) + Integer(2)*z12**Integer(4) +
↪ Integer(3)*z12**Integer(3) + z12**Integer(2) + Integer(2)*z12
>>> phi = DrinfeldModule(A, [p_root, z12**Integer(3), z12**Integer(5)])
>>> phi.characteristic()
T^2 + (4*z2 + 2)*T + 2
>>> phi.base_morphism()(phi.characteristic())
0
```

```
sage: B.<Y> = Fq[]
sage: L = Frac(B)
sage: psi = DrinfeldModule(A, [L(1), 0, 0, L(1)])
sage: psi.characteristic()
Traceback (most recent call last):
...
NotImplementedError: function ring characteristic not implemented in this
↪ case
```

```
>>> from sage.all import *
>>> B = Fq['Y']; (Y,) = B._first_ngens(1)
>>> L = Frac(B)
>>> psi = DrinfeldModule(A, [L(Integer(1)), Integer(0), Integer(0),
↪ L(Integer(1))])
>>> psi.characteristic()
Traceback (most recent call last):
...
NotImplementedError: function ring characteristic not implemented in this
↪ case
```

constant_coefficient()

Return the constant coefficient of the generator of this Drinfeld module.

OUTPUT: an element in the base field

EXAMPLES:

```

sage: Fq = GF(25)
sage: A.<T> = Fq[]
sage: K.<z12> = Fq.extension(6)
sage: p_root = 2*z12^11 + 2*z12^10 + z12^9 + 3*z12^8 + z12^7 + 2*z12^5 +
↳ 2*z12^4 + 3*z12^3 + z12^2 + 2*z12
sage: phi = DrinfeldModule(A, [p_root, z12^3, z12^5])
sage: phi.constant_coefficient() == p_root
True

```

```

>>> from sage.all import *
>>> Fq = GF(Integer(25))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(6), names=('z12',)); (z12,) = K._first_
↳ ngens(1)
>>> p_root = Integer(2)*z12**Integer(11) + Integer(2)*z12**Integer(10) +
↳ z12**Integer(9) + Integer(3)*z12**Integer(8) + z12**Integer(7) +
↳ Integer(2)*z12**Integer(5) + Integer(2)*z12**Integer(4) +
↳ Integer(3)*z12**Integer(3) + z12**Integer(2) + Integer(2)*z12
>>> phi = DrinfeldModule(A, [p_root, z12**Integer(3), z12**Integer(5)])
>>> phi.constant_coefficient() == p_root
True

```

Let $\mathbb{F}_q[T]$ be the function ring, and let γ be the base of the Drinfeld module. The constant coefficient is $\gamma(T)$:

```

sage: C = phi.category()
sage: base = C.base()
sage: base(T) == phi.constant_coefficient()
True

```

```

>>> from sage.all import *
>>> C = phi.category()
>>> base = C.base()
>>> base(T) == phi.constant_coefficient()
True

```

Naturally, two Drinfeld modules in the same category have the same constant coefficient:

```

sage: t = phi.ore_polring().gen()
sage: psi = C.object(phi.constant_coefficient() + t^3)
sage: psi
Drinfeld module defined by T |--> t^3 + 2*z12^11 + 2*z12^10 + z12^9 +
↳ 3*z12^8 + z12^7 + 2*z12^5 + 2*z12^4 + 3*z12^3 + z12^2 + 2*z12

```

```

>>> from sage.all import *
>>> t = phi.ore_polring().gen()
>>> psi = C.object(phi.constant_coefficient() + t**Integer(3))
>>> psi
Drinfeld module defined by T |--> t^3 + 2*z12^11 + 2*z12^10 + z12^9 +
↳ 3*z12^8 + z12^7 + 2*z12^5 + 2*z12^4 + 3*z12^3 + z12^2 + 2*z12

```

Reciprocally, it is impossible to create two Drinfeld modules in this category if they do not share the same constant coefficient:

```

sage: rho = C.object(phi.constant_coefficient() + 1 + t^3)
Traceback (most recent call last):

```

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```
...
ValueError: constant coefficient must equal that of the category
```

```
>>> from sage.all import *
>>> rho = C.object(phi.constant_coefficient() + Integer(1) +_
↳ t**Integer(3))
Traceback (most recent call last):
...
ValueError: constant coefficient must equal that of the category
```

function_ring()

Return the function ring of this Drinfeld module.

EXAMPLES:

```
sage: Fq = GF(25)
sage: A.<T> = Fq[]
sage: K.<z12> = Fq.extension(6)
sage: p_root = 2*z12^11 + 2*z12^10 + z12^9 + 3*z12^8 + z12^7 + 2*z12^5 +_
↳ 2*z12^4 + 3*z12^3 + z12^2 + 2*z12
sage: phi = DrinfeldModule(A, [p_root, z12^3, z12^5])
sage: phi.function_ring()
Univariate Polynomial Ring in T over Finite Field in z2 of size 5^2
sage: phi.function_ring() is A
True
```

```
>>> from sage.all import *
>>> Fq = GF(Integer(25))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(6), names=('z12',)); (z12,) = K._first_
↳ ngens(1)
>>> p_root = Integer(2)*z12**Integer(11) + Integer(2)*z12**Integer(10) +_
↳ z12**Integer(9) + Integer(3)*z12**Integer(8) + z12**Integer(7) +_
↳ Integer(2)*z12**Integer(5) + Integer(2)*z12**Integer(4) +_
↳ Integer(3)*z12**Integer(3) + z12**Integer(2) + Integer(2)*z12
>>> phi = DrinfeldModule(A, [p_root, z12**Integer(3), z12**Integer(5)])
>>> phi.function_ring()
Univariate Polynomial Ring in T over Finite Field in z2 of size 5^2
>>> phi.function_ring() is A
True
```

ore_polring()

Return the Ore polynomial ring of this Drinfeld module.

EXAMPLES:

```
sage: Fq = GF(25)
sage: A.<T> = Fq[]
sage: K.<z12> = Fq.extension(6)
sage: p_root = 2*z12^11 + 2*z12^10 + z12^9 + 3*z12^8 + z12^7 + 2*z12^5 +_
↳ 2*z12^4 + 3*z12^3 + z12^2 + 2*z12
sage: phi = DrinfeldModule(A, [p_root, z12^3, z12^5])
sage: S = phi.ore_polring()
sage: S
Ore Polynomial Ring in t over Finite Field in z12 of size 5^12 over its_
↳ base twisted by Frob^2
```

```

>>> from sage.all import *
>>> Fq = GF(Integer(25))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(6), names=('z12',)); (z12,) = K._first_
↳ngens(1)
>>> p_root = Integer(2)*z12**Integer(11) + Integer(2)*z12**Integer(10) +
↳z12**Integer(9) + Integer(3)*z12**Integer(8) + z12**Integer(7) +
↳Integer(2)*z12**Integer(5) + Integer(2)*z12**Integer(4) +
↳Integer(3)*z12**Integer(3) + z12**Integer(2) + Integer(2)*z12
>>> phi = DrinfeldModule(A, [p_root, z12**Integer(3), z12**Integer(5)])
>>> S = phi.ore_polring()
>>> S
Ore Polynomial Ring in t over Finite Field in z12 of size 5^12 over its
↳base twisted by Frob^2

```

The Ore polynomial ring can also be retrieved from the category of the Drinfeld module:

```

sage: S is phi.category().ore_polring()
True

```

```

>>> from sage.all import *
>>> S is phi.category().ore_polring()
True

```

The generator of the Drinfeld module is in the Ore polynomial ring:

```

sage: phi(T) in S
True

```

```

>>> from sage.all import *
>>> phi(T) in S
True

```

ore_variable()

Return the variable of the Ore polynomial ring of this Drinfeld module.

EXAMPLES:

```

sage: Fq = GF(25)
sage: A.<T> = Fq[]
sage: K.<z12> = Fq.extension(6)
sage: p_root = 2*z12^11 + 2*z12^10 + z12^9 + 3*z12^8 + z12^7 + 2*z12^5 +
↳2*z12^4 + 3*z12^3 + z12^2 + 2*z12
sage: phi = DrinfeldModule(A, [p_root, z12^3, z12^5])

sage: phi.ore_polring()
Ore Polynomial Ring in t over Finite Field in z12 of size 5^12 over its
↳base twisted by Frob^2
sage: phi.ore_variable()
t

```

```

>>> from sage.all import *
>>> Fq = GF(Integer(25))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(6), names=('z12',)); (z12,) = K._first_
↳ngens(1)

```

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```

>>> p_root = Integer(2)*z12**Integer(11) + Integer(2)*z12**Integer(10) +
↳ z12**Integer(9) + Integer(3)*z12**Integer(8) + z12**Integer(7) +
↳ Integer(2)*z12**Integer(5) + Integer(2)*z12**Integer(4) +
↳ Integer(3)*z12**Integer(3) + z12**Integer(2) + Integer(2)*z12
>>> phi = DrinfeldModule(A, [p_root, z12**Integer(3), z12**Integer(5)])

>>> phi.ore_polring()
Ore Polynomial Ring in t over Finite Field in z12 of size 5^12 over its
↳ base twisted by Frob^2
>>> phi.ore_variable()
t

```

base_morphism()

Return the base morphism of the category.

EXAMPLES:

```

sage: Fq = GF(11)
sage: A.<T> = Fq[]
sage: K.<z> = Fq.extension(4)
sage: p_root = z^3 + 7*z^2 + 6*z + 10
sage: phi = DrinfeldModule(A, [p_root, 0, 0, 1])
sage: C = phi.category()
sage: C.base_morphism()
Ring morphism:
  From: Univariate Polynomial Ring in T over Finite Field of size 11
  To:   Finite Field in z of size 11^4 over its base
  Defn: T |--> z^3 + 7*z^2 + 6*z + 10

sage: C.constant_coefficient() == C.base_morphism()(T)
True

```

```

>>> from sage.all import *
>>> Fq = GF(Integer(11))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(4), names=('z',)); (z,) = K._first_ngens(1)
>>> p_root = z**Integer(3) + Integer(7)*z**Integer(2) + Integer(6)*z +
↳ Integer(10)
>>> phi = DrinfeldModule(A, [p_root, Integer(0), Integer(0), Integer(1)])
>>> C = phi.category()
>>> C.base_morphism()
Ring morphism:
  From: Univariate Polynomial Ring in T over Finite Field of size 11
  To:   Finite Field in z of size 11^4 over its base
  Defn: T |--> z^3 + 7*z^2 + 6*z + 10

>>> C.constant_coefficient() == C.base_morphism()(T)
True

```

base_over_constants_field()Return the base field, seen as an extension over the constants field $\mathbb{F}_q$.

EXAMPLES:

```

sage: Fq = GF(11)
sage: A.<T> = Fq[]

```

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```

sage: K.<z> = Fq.extension(4)
sage: p_root = z^3 + 7*z^2 + 6*z + 10
sage: phi = DrinfeldModule(A, [p_root, 0, 0, 1])
sage: C = phi.category()
sage: C.base_over_constants_field()
Field in z with defining polynomial x^4 + 8*x^2 + 10*x + 2 over its base

```

```

>>> from sage.all import *
>>> Fq = GF(Integer(11))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(4), names=('z',)); (z,) = K._first_ngens(1)
>>> p_root = z**Integer(3) + Integer(7)*z**Integer(2) + Integer(6)*z +
↪Integer(10)
>>> phi = DrinfeldModule(A, [p_root, Integer(0), Integer(0), Integer(1)])
>>> C = phi.category()
>>> C.base_over_constants_field()
Field in z with defining polynomial x^4 + 8*x^2 + 10*x + 2 over its base

```

characteristic()

Return the function ring-characteristic.

EXAMPLES:

```

sage: Fq = GF(11)
sage: A.<T> = Fq[]
sage: K.<z> = Fq.extension(4)
sage: p_root = z^3 + 7*z^2 + 6*z + 10
sage: phi = DrinfeldModule(A, [p_root, 0, 0, 1])
sage: C = phi.category()
sage: C.characteristic()
T^2 + 7*T + 2

```

```

>>> from sage.all import *
>>> Fq = GF(Integer(11))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(4), names=('z',)); (z,) = K._first_ngens(1)
>>> p_root = z**Integer(3) + Integer(7)*z**Integer(2) + Integer(6)*z +
↪Integer(10)
>>> phi = DrinfeldModule(A, [p_root, Integer(0), Integer(0), Integer(1)])
>>> C = phi.category()
>>> C.characteristic()
T^2 + 7*T + 2

```

```

sage: psi = DrinfeldModule(A, [Frac(A).gen(), 1])
sage: C = psi.category()
sage: C.characteristic()
0

```

```

>>> from sage.all import *
>>> psi = DrinfeldModule(A, [Frac(A).gen(), Integer(1)])
>>> C = psi.category()
>>> C.characteristic()
0

```

constant_coefficient()

Return the constant coefficient of the category.

EXAMPLES:

```

sage: Fq = GF(11)
sage: A.<T> = Fq[]
sage: K.<z> = Fq.extension(4)
sage: p_root = z^3 + 7*z^2 + 6*z + 10
sage: phi = DrinfeldModule(A, [p_root, 0, 0, 1])
sage: C = phi.category()
sage: C.constant_coefficient()
z^3 + 7*z^2 + 6*z + 10
sage: C.constant_coefficient() == C.base()(T)
True

```

```

>>> from sage.all import *
>>> Fq = GF(Integer(11))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(4), names=('z',)); (z,) = K._first_ngens(1)
>>> p_root = z**Integer(3) + Integer(7)*z**Integer(2) + Integer(6)*z +
↳ Integer(10)
>>> phi = DrinfeldModule(A, [p_root, Integer(0), Integer(0), Integer(1)])
>>> C = phi.category()
>>> C.constant_coefficient()
z^3 + 7*z^2 + 6*z + 10
>>> C.constant_coefficient() == C.base()(T)
True

```

function_ring()

Return the function ring of the category.

EXAMPLES:

```

sage: Fq = GF(11)
sage: A.<T> = Fq[]
sage: K.<z> = Fq.extension(4)
sage: p_root = z^3 + 7*z^2 + 6*z + 10
sage: phi = DrinfeldModule(A, [p_root, 0, 0, 1])
sage: C = phi.category()
sage: C.function_ring()
Univariate Polynomial Ring in T over Finite Field of size 11
sage: C.function_ring() is A
True

```

```

>>> from sage.all import *
>>> Fq = GF(Integer(11))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(4), names=('z',)); (z,) = K._first_ngens(1)
>>> p_root = z**Integer(3) + Integer(7)*z**Integer(2) + Integer(6)*z +
↳ Integer(10)
>>> phi = DrinfeldModule(A, [p_root, Integer(0), Integer(0), Integer(1)])
>>> C = phi.category()
>>> C.function_ring()
Univariate Polynomial Ring in T over Finite Field of size 11
>>> C.function_ring() is A
True

```

object (gen)

Return a Drinfeld module object in the category whose generator is the input.

INPUT:

- `gen` – the generator of the Drinfeld module, given as an Ore polynomial or a list of coefficients

EXAMPLES:

```
sage: Fq = GF(11)
sage: A.<T> = Fq[]
sage: K.<z> = Fq.extension(4)
sage: p_root = z^3 + 7*z^2 + 6*z + 10
sage: psi = DrinfeldModule(A, [p_root, 1])
sage: C = psi.category()

sage: phi = C.object([p_root, 0, 1])
sage: phi
Drinfeld module defined by T |--> t^2 + z^3 + 7*z^2 + 6*z + 10
sage: t = phi.ore_polring().gen()
sage: C.object(t^2 + z^3 + 7*z^2 + 6*z + 10) is phi
True
```

```
>>> from sage.all import *
>>> Fq = GF(Integer(11))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(4), names=('z',)); (z,) = K._first_ngens(1)
>>> p_root = z**Integer(3) + Integer(7)*z**Integer(2) + Integer(6)*z +
↳ Integer(10)
>>> psi = DrinfeldModule(A, [p_root, Integer(1)])
>>> C = psi.category()

>>> phi = C.object([p_root, Integer(0), Integer(1)])
>>> phi
Drinfeld module defined by T |--> t^2 + z^3 + 7*z^2 + 6*z + 10
>>> t = phi.ore_polring().gen()
>>> C.object(t**Integer(2) + z**Integer(3) + Integer(7)*z**Integer(2) +
↳ Integer(6)*z + Integer(10)) is phi
True
```

`ore_polring()`

Return the Ore polynomial ring of the category

EXAMPLES:

```
sage: Fq = GF(11)
sage: A.<T> = Fq[]
sage: K.<z> = Fq.extension(4)
sage: p_root = z^3 + 7*z^2 + 6*z + 10
sage: phi = DrinfeldModule(A, [p_root, 0, 0, 1])
sage: C = phi.category()
sage: C.ore_polring()
Ore Polynomial Ring in t over Finite Field in z of size 11^4 over its base
↳ twisted by Frob
```

```
>>> from sage.all import *
>>> Fq = GF(Integer(11))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(4), names=('z',)); (z,) = K._first_ngens(1)
>>> p_root = z**Integer(3) + Integer(7)*z**Integer(2) + Integer(6)*z +
↳ Integer(10)
```

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```

>>> phi = DrinfeldModule(A, [p_root, Integer(0), Integer(0), Integer(1)])
>>> C = phi.category()
>>> C.ore_polring()
Ore Polynomial Ring in t over Finite Field in z of size 11^4 over its base
↳ twisted by Frob

```

random_object (rank)

Return a random Drinfeld module in the category with given rank.

INPUT:

- rank – an integer, the rank of the Drinfeld module

EXAMPLES:

```

sage: Fq = GF(11)
sage: A.<T> = Fq[]
sage: K.<z> = Fq.extension(4)
sage: p_root = z^3 + 7*z^2 + 6*z + 10
sage: phi = DrinfeldModule(A, [p_root, 0, 0, 1])
sage: C = phi.category()

sage: psi = C.random_object(3) # random
Drinfeld module defined by T |--> (6*z^3 + 4*z^2 + 10*z + 9)*t^3 + (4*z^3 +
↳ 8*z^2 + 8*z)*t^2 + (10*z^3 + 3*z^2 + 6*z)*t + z^3 + 7*z^2 + 6*z + 10
sage: psi.rank() == 3
True

```

```

>>> from sage.all import *
>>> Fq = GF(Integer(11))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(4), names=('z',)); (z,) = K._first_ngens(1)
>>> p_root = z**Integer(3) + Integer(7)*z**Integer(2) + Integer(6)*z +
↳ Integer(10)
>>> phi = DrinfeldModule(A, [p_root, Integer(0), Integer(0), Integer(1)])
>>> C = phi.category()

>>> psi = C.random_object(Integer(3)) # random
Drinfeld module defined by T |--> (6*z^3 + 4*z^2 + 10*z + 9)*t^3 + (4*z^3 +
↳ 8*z^2 + 8*z)*t^2 + (10*z^3 + 3*z^2 + 6*z)*t + z^3 + 7*z^2 + 6*z + 10
>>> psi.rank() == Integer(3)
True

```

super_categories ()

EXAMPLES:

```

sage: Fq = GF(11)
sage: A.<T> = Fq[]
sage: K.<z> = Fq.extension(4)
sage: p_root = z^3 + 7*z^2 + 6*z + 10
sage: phi = DrinfeldModule(A, [p_root, 0, 0, 1])
sage: C = phi.category()
sage: C.super_categories()
[Category of objects]

```

```
>>> from sage.all import *
>>> Fq = GF(Integer(11))
>>> A = Fq['T']; (T,) = A._first_ngens(1)
>>> K = Fq.extension(Integer(4), names=('z',)); (z,) = K._first_ngens(1)
>>> p_root = z**Integer(3) + Integer(7)*z**Integer(2) + Integer(6)*z + Integer(10)
>>> phi = DrinfeldModule(A, [p_root, Integer(0), Integer(0), Integer(1)])
>>> C = phi.category()
>>> C.super_categories()
[Category of objects]
```


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